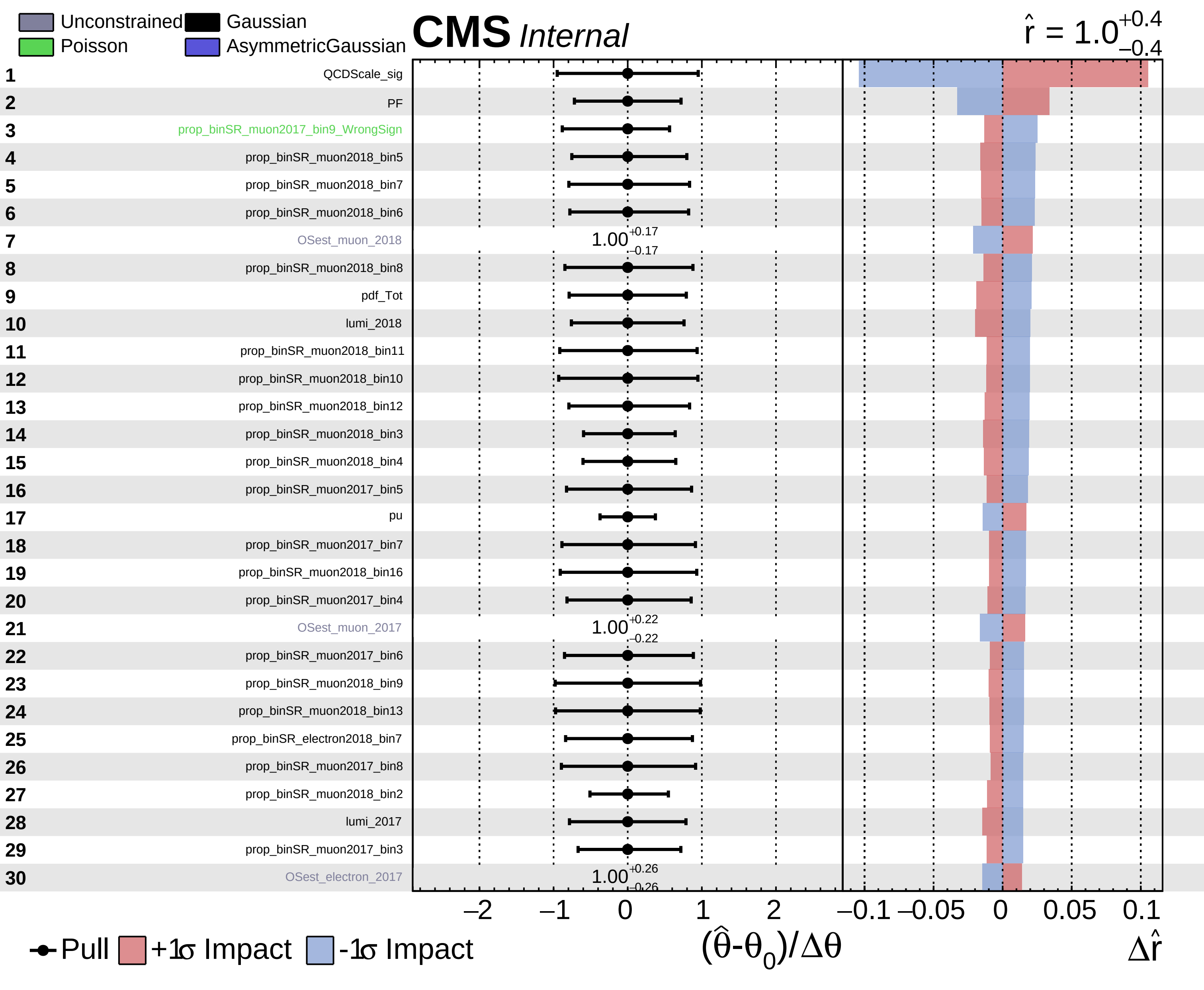
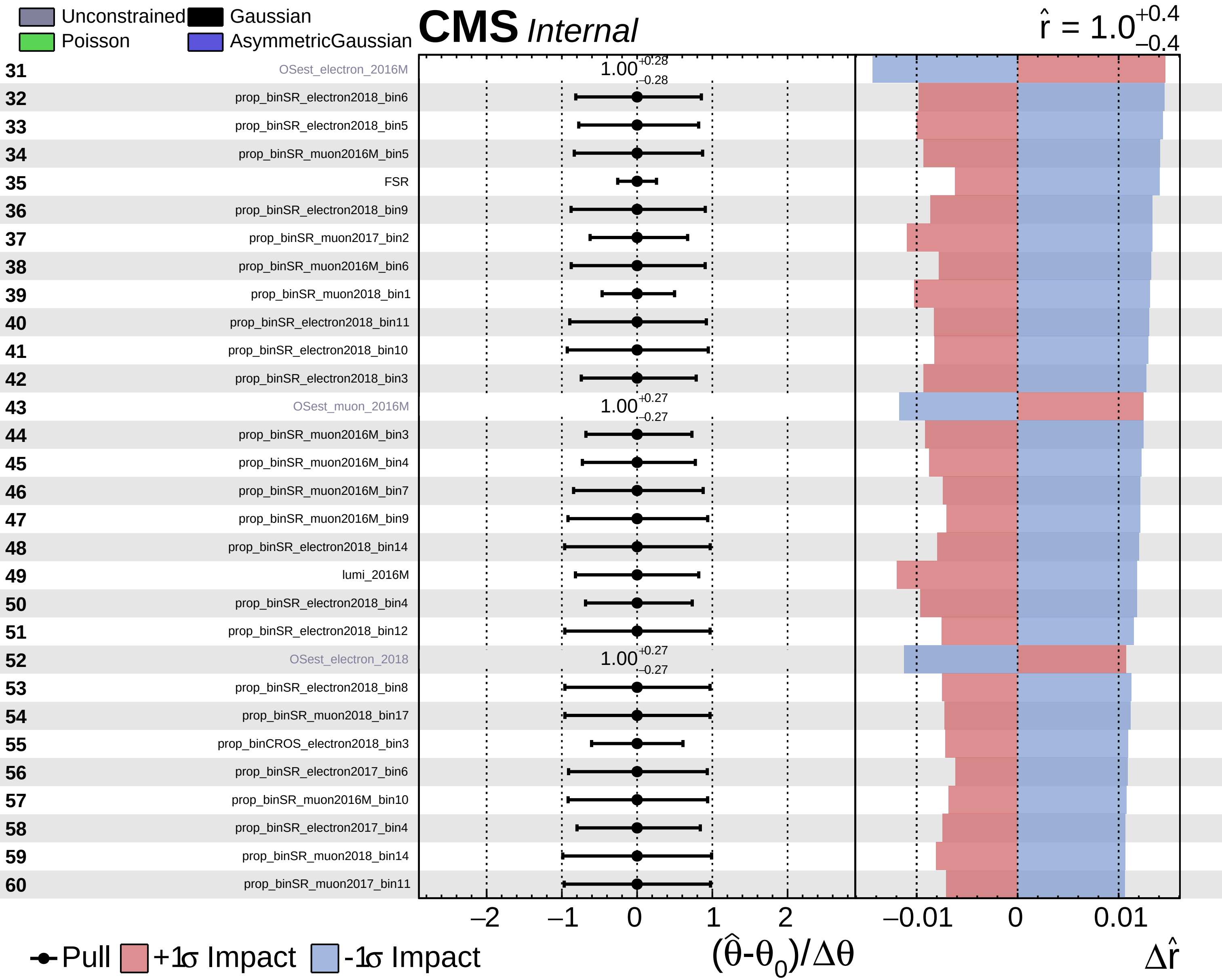
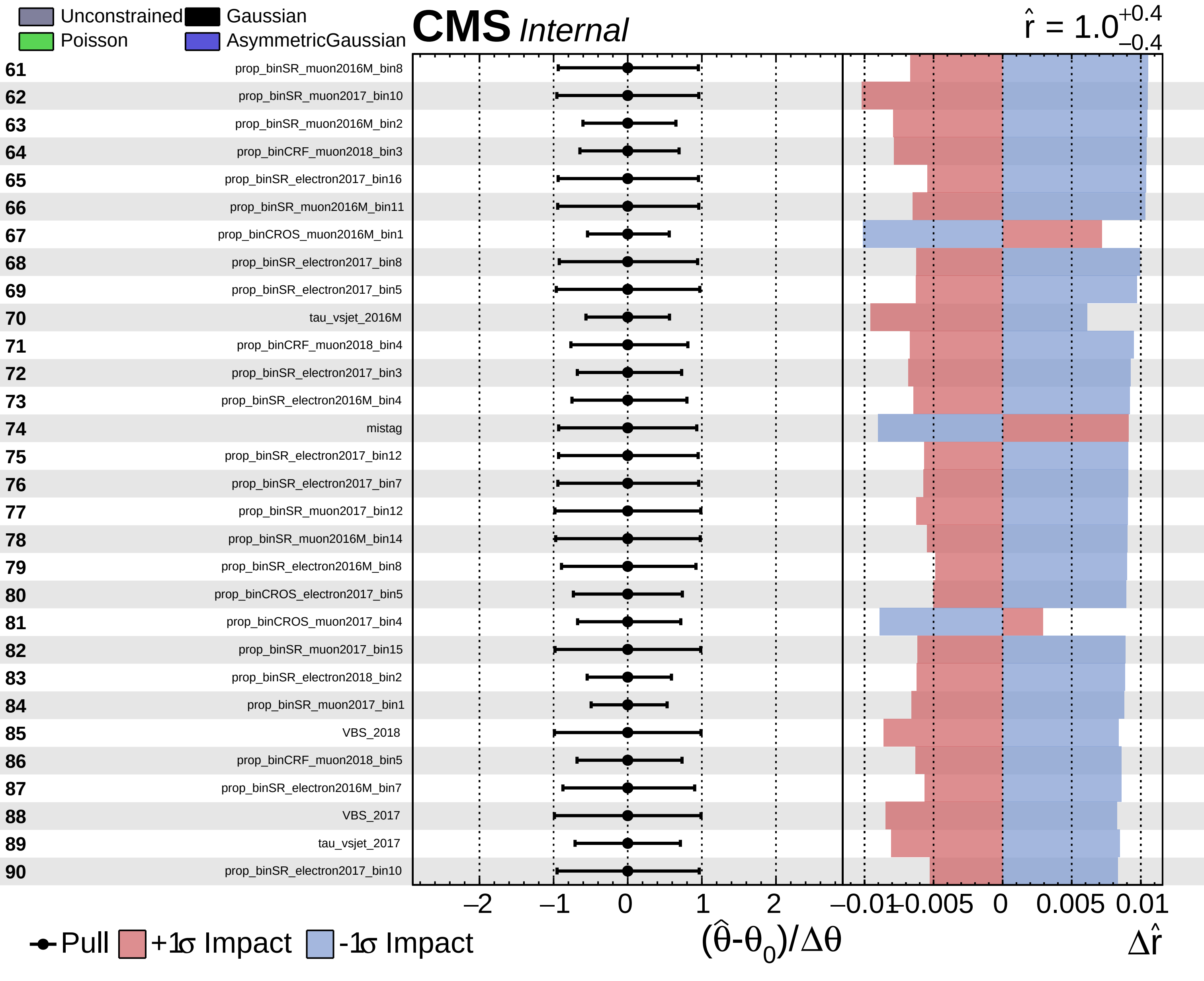


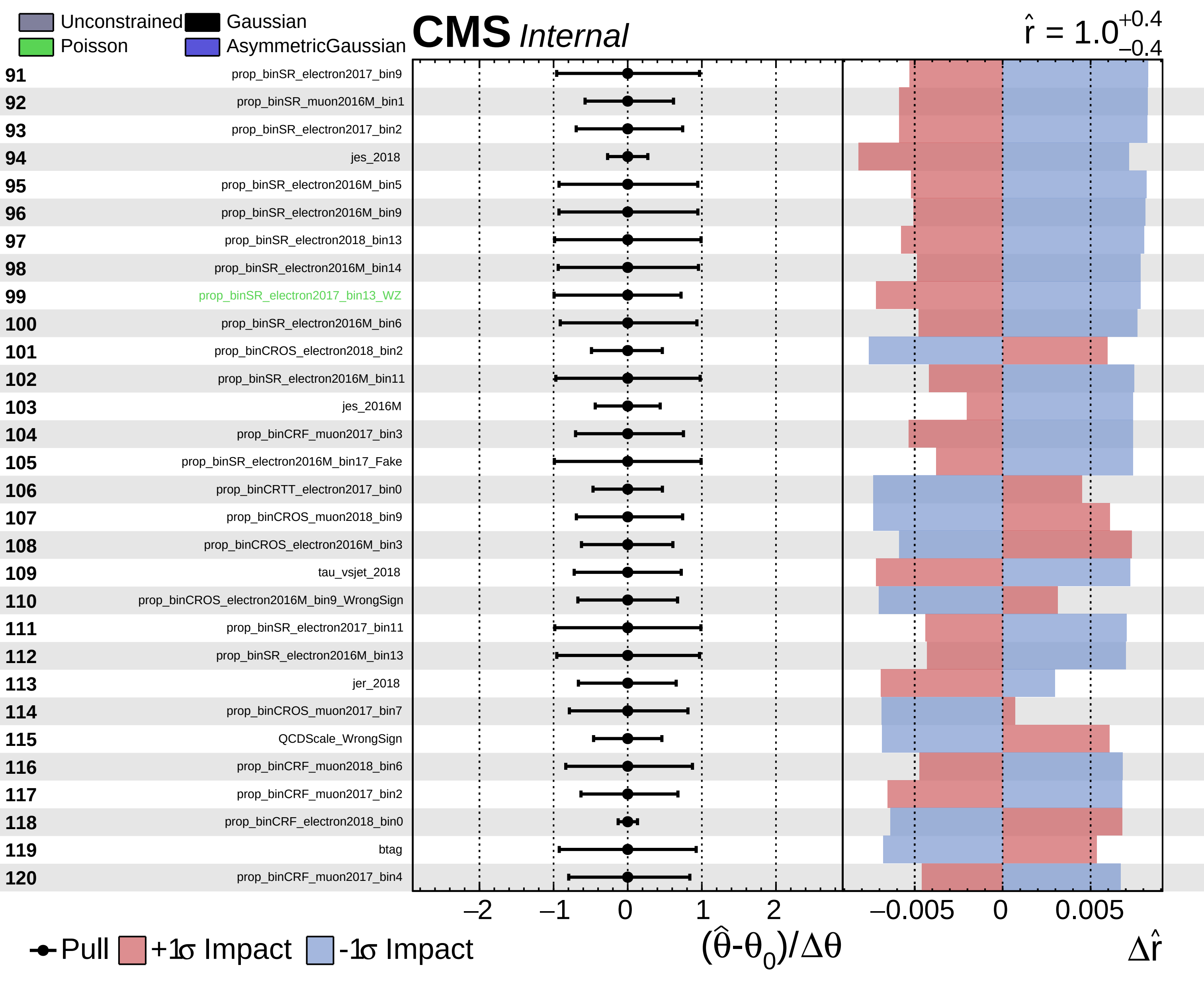
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 1.0 +0.4 −0.4
1 QCDScale_sig
2 PF
3 prop_binSR_muon2017_bin9_WrongSign
4 prop_binSR_muon2018_bin5
5 prop_binSR_muon2018_bin7
6 prop_binSR_muon2018_bin6
7 OSest_muon_2018
1.00 +0.17 −0.17
8 prop_binSR_muon2018_bin8
9 pdf_Tot
10 lumi_2018
11 prop_binSR_muon2018_bin11
12 prop_binSR_muon2018_bin10
13 prop_binSR_muon2018_bin12
14 prop_binSR_muon2018_bin3
15 prop_binSR_muon2018_bin4
16 prop_binSR_muon2017_bin5
17 pu
18 prop_binSR_muon2017_bin7
19 prop_binSR_muon2018_bin16
20 prop_binSR_muon2017_bin4
21 OSest_muon_2017
1.00 +0.22 −0.22
22 prop_binSR_muon2017_bin6
23 prop_binSR_muon2018_bin9
24 prop_binSR_muon2018_bin13
25 prop_binSR_electron2018_bin7
26 prop_binSR_muon2017_bin8
27 prop_binSR_muon2018_bin2
28 lumi_2017
29 prop_binSR_muon2017_bin3
30 OSest_electron_2017
1.00 +0.26 −0.26
−2
−1
0
1
2
−0.1
−0.05
0
0.05
0.1
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂



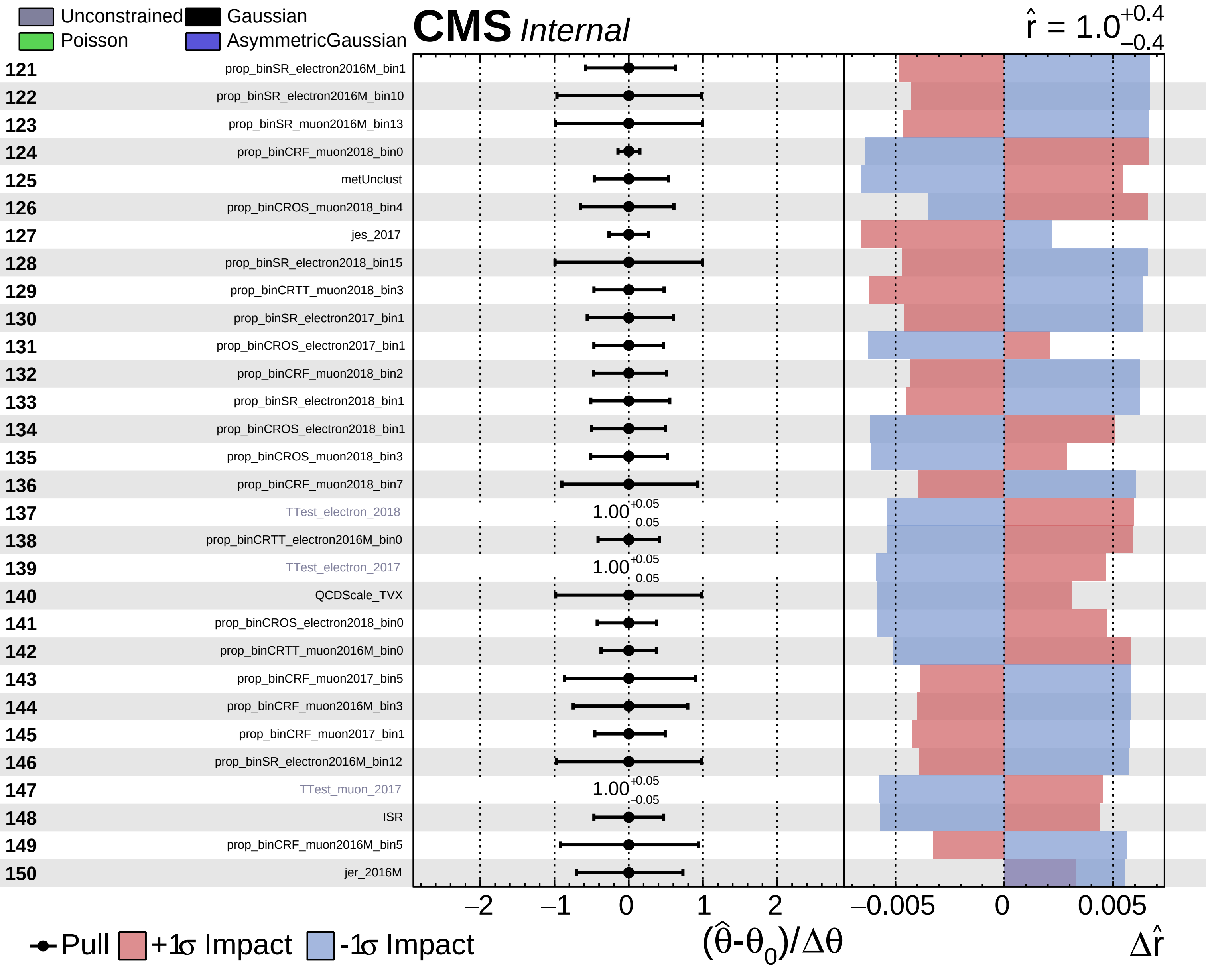
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 1.0 +0.4 −0.4
31 OSest_electron_2016M 1.00 +0.28 −0.28
32 prop_binSR_electron2018_bin6
33 prop_binSR_electron2018_bin5
34 prop_binSR_muon2016M_bin5
35 FSR
36 prop_binSR_electron2018_bin9
37 prop_binSR_muon2017_bin2
38 prop_binSR_muon2016M_bin6
39 prop_binSR_muon2018_bin1
40 prop_binSR_electron2018_bin11
41 prop_binSR_electron2018_bin10
42 prop_binSR_electron2018_bin3
43 OSest_muon_2016M 1.00 +0.27 −0.27
44 prop_binSR_muon2016M_bin3
45 prop_binSR_muon2016M_bin4
46 prop_binSR_muon2016M_bin7
47 prop_binSR_muon2016M_bin9
48 prop_binSR_electron2018_bin14
49 lumi_2016M
50 prop_binSR_electron2018_bin4
51 prop_binSR_electron2018_bin12
52 OSest_electron_2018 1.00 +0.27 −0.27
53 prop_binSR_electron2018_bin8
54 prop_binSR_muon2018_bin17
55 prop_binCROS_electron2018_bin3
56 prop_binSR_electron2017_bin6
57 prop_binSR_muon2016M_bin10
58 prop_binSR_electron2017_bin4
59 prop_binSR_muon2018_bin14
60 prop_binSR_muon2017_bin11
−2 −1 0 1 2
−0.01 0 0.01
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂



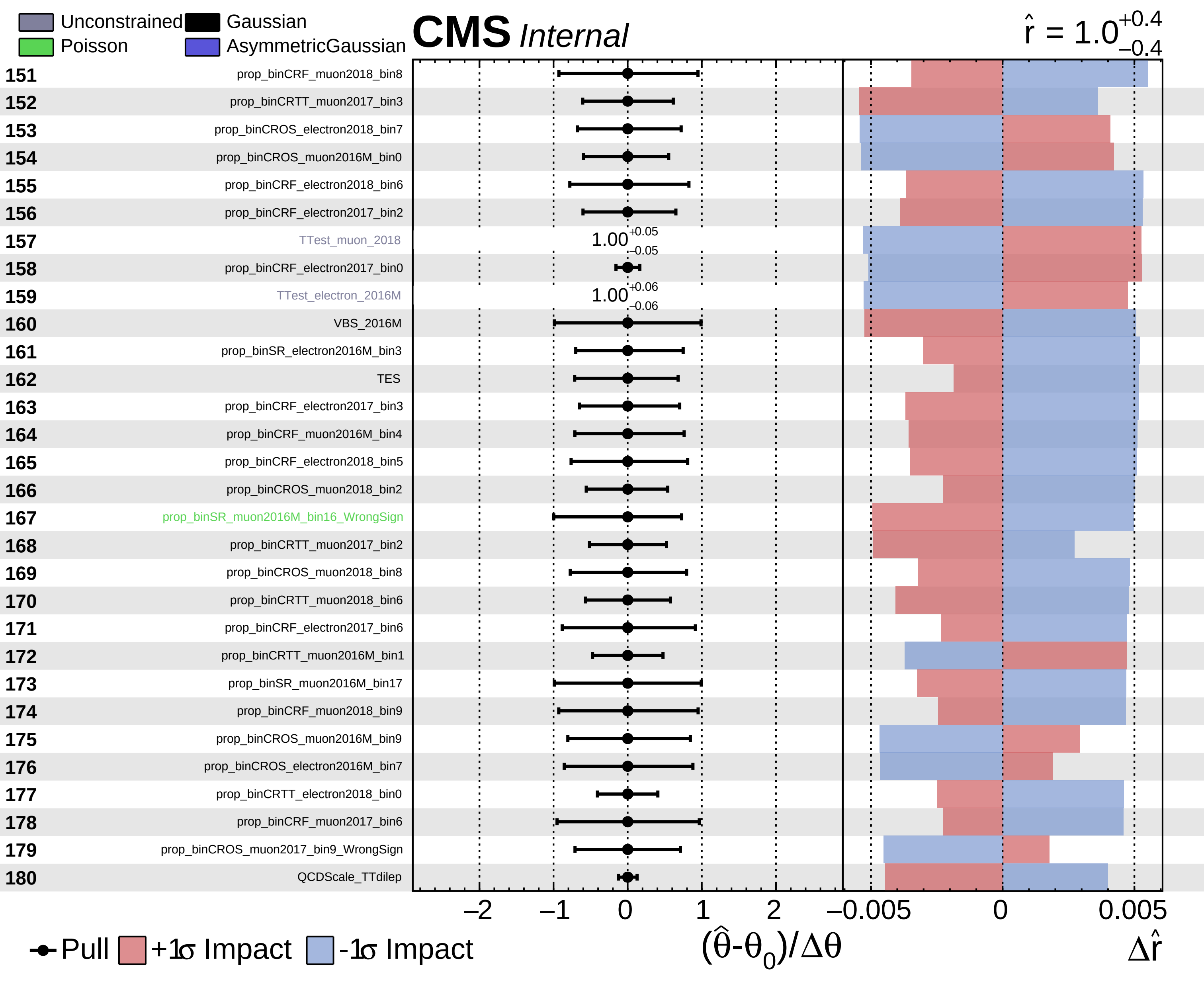
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.4}_{-0.4}$
61 prop_binSR_muon2016M_bin8
62 prop_binSR_muon2017_bin10
63 prop_binSR_muon2016M_bin2
64 prop_binCRF_muon2018_bin3
65 prop_binSR_electron2017_bin16
66 prop_binSR_muon2016M_bin11
67 prop_binCROS_muon2016M_bin1
68 prop_binSR_electron2017_bin8
69 prop_binSR_electron2017_bin5
70 tau_vsjet_2016M
71 prop_binCRF_muon2018_bin4
72 prop_binSR_electron2017_bin3
73 prop_binSR_electron2016M_bin4
74 mistag
75 prop_binSR_electron2017_bin12
76 prop_binSR_electron2017_bin7
77 prop_binSR_muon2017_bin12
78 prop_binSR_muon2016M_bin14
79 prop_binSR_electron2016M_bin8
80 prop_binCROS_electron2017_bin5
81 prop_binCROS_muon2017_bin4
82 prop_binSR_muon2017_bin15
83 prop_binSR_electron2018_bin2
84 prop_binSR_muon2017_bin1
85 VBS_2018
86 prop_binCRF_muon2018_bin5
87 prop_binSR_electron2016M_bin7
88 VBS_2017
89 tau_vsjet_2017
90 prop_binSR_electron2017_bin10
−2 −1 0 1 2
−0.01 −0.005 0 0.005 0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



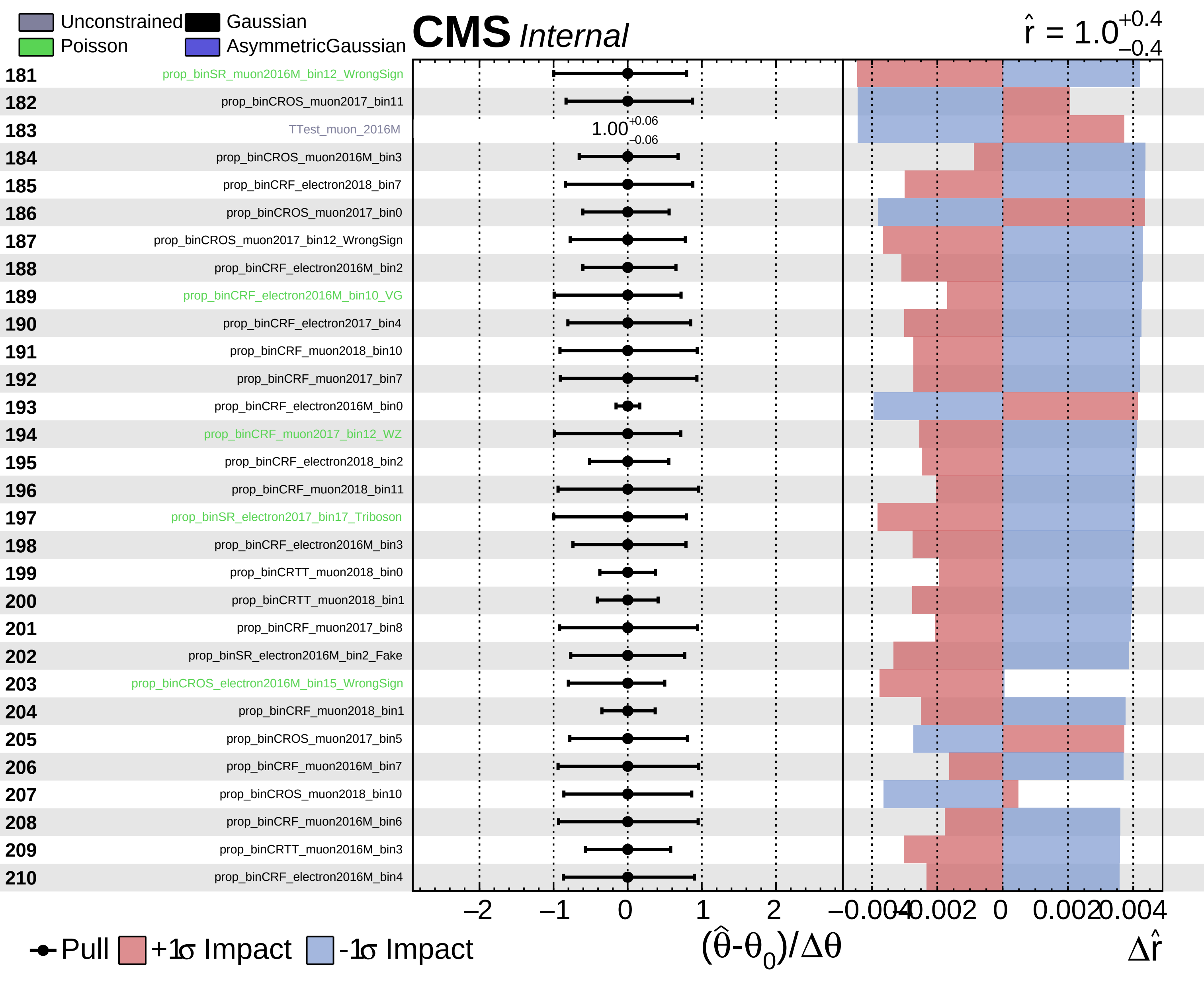
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.4}_{-0.4}$
91 prop_binSR_electron2017_bin9
92 prop_binSR_muon2016M_bin1
93 prop_binSR_electron2017_bin2
94 jes_2018
95 prop_binSR_electron2016M_bin5
96 prop_binSR_electron2016M_bin9
97 prop_binSR_electron2018_bin13
98 prop_binSR_electron2016M_bin14
99 prop_binSR_electron2017_bin13_WZ
100 prop_binSR_electron2016M_bin6
101 prop_binCROS_electron2018_bin2
102 prop_binSR_electron2016M_bin11
103 jes_2016M
104 prop_binCRF_muon2017_bin3
105 prop_binSR_electron2016M_bin17_Fake
106 prop_binCRTT_electron2017_bin0
107 prop_binCROS_muon2018_bin9
108 prop_binCROS_electron2016M_bin3
109 tau_vsjet_2018
110 prop_binCROS_electron2016M_bin9_WrongSign
111 prop_binSR_electron2017_bin11
112 prop_binSR_electron2016M_bin13
113 jer_2018
114 prop_binCROS_muon2017_bin7
115 QCDScale_WrongSign
116 prop_binCRF_muon2018_bin6
117 prop_binCRF_muon2017_bin2
118 prop_binCRF_electron2018_bin0
119 btag
120 prop_binCRF_muon2017_bin4
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



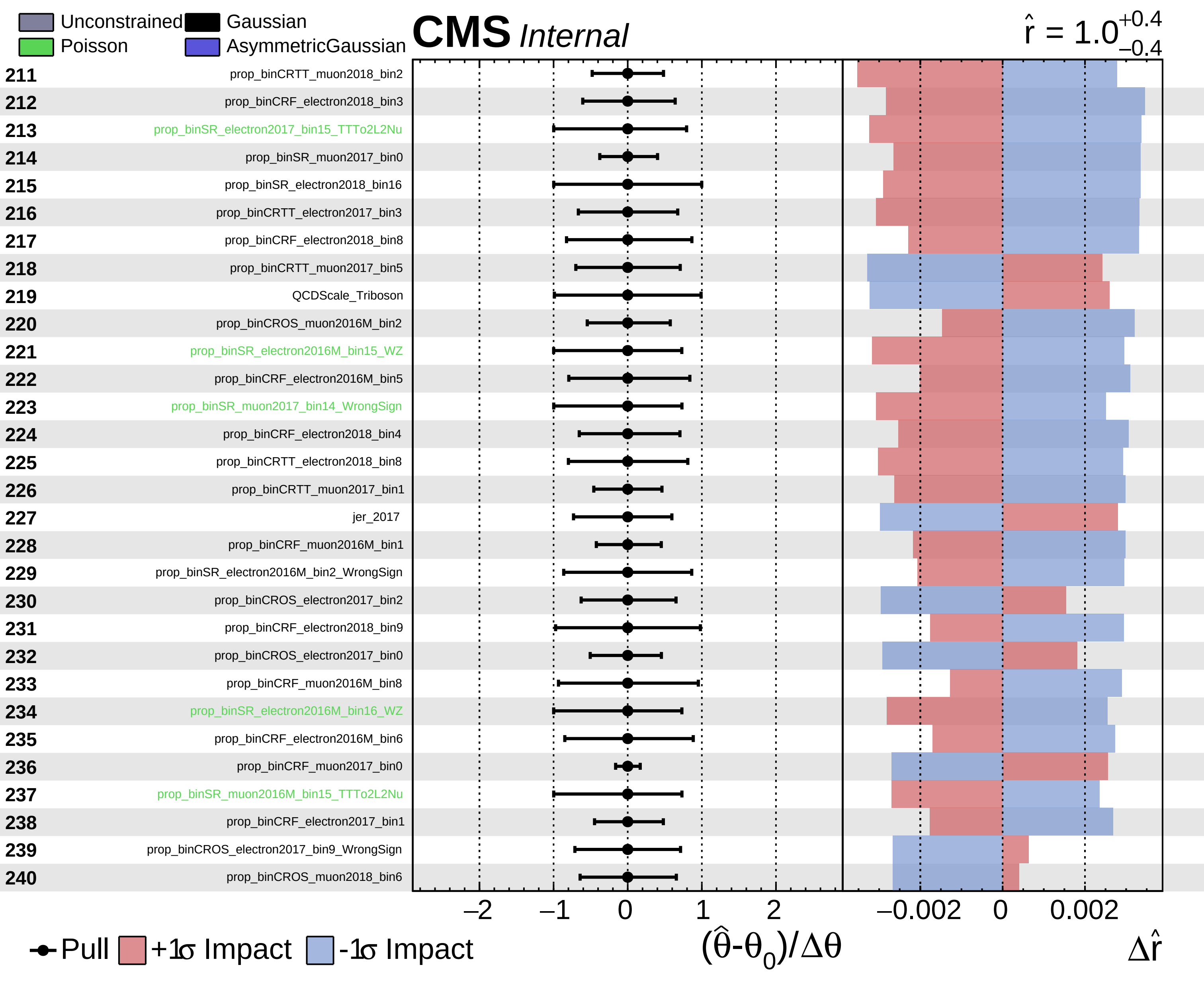
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.4}_{-0.4}$
121 prop_binSR_electron2016M_bin1
122 prop_binSR_electron2016M_bin10
123 prop_binSR_muon2016M_bin13
124 prop_binCRF_muon2018_bin0
125 metUnclust
126 prop_binCROS_muon2018_bin4
127 jes_2017
128 prop_binSR_electron2018_bin15
129 prop_binCRTT_muon2018_bin3
130 prop_binSR_electron2017_bin1
131 prop_binCROS_electron2017_bin1
132 prop_binCRF_muon2018_bin2
133 prop_binSR_electron2018_bin1
134 prop_binCROS_electron2018_bin1
135 prop_binCROS_muon2018_bin3
136 prop_binCRF_muon2018_bin7
137 TTest_electron_2018
$1.00^{+0.05}_{-0.05}$
138 prop_binCRTT_electron2016M_bin0
139 TTest_electron_2017
$1.00^{+0.05}_{-0.05}$
140 QCDScale_TVX
141 prop_binCROS_electron2018_bin0
142 prop_binCRTT_muon2016M_bin0
143 prop_binCRF_muon2017_bin5
144 prop_binCRF_muon2016M_bin3
145 prop_binCRF_muon2017_bin1
146 prop_binSR_electron2016M_bin12
147 TTest_muon_2017
$1.00^{+0.05}_{-0.05}$
148 ISR
149 prop_binCRF_muon2016M_bin5
150 jer_2016M
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

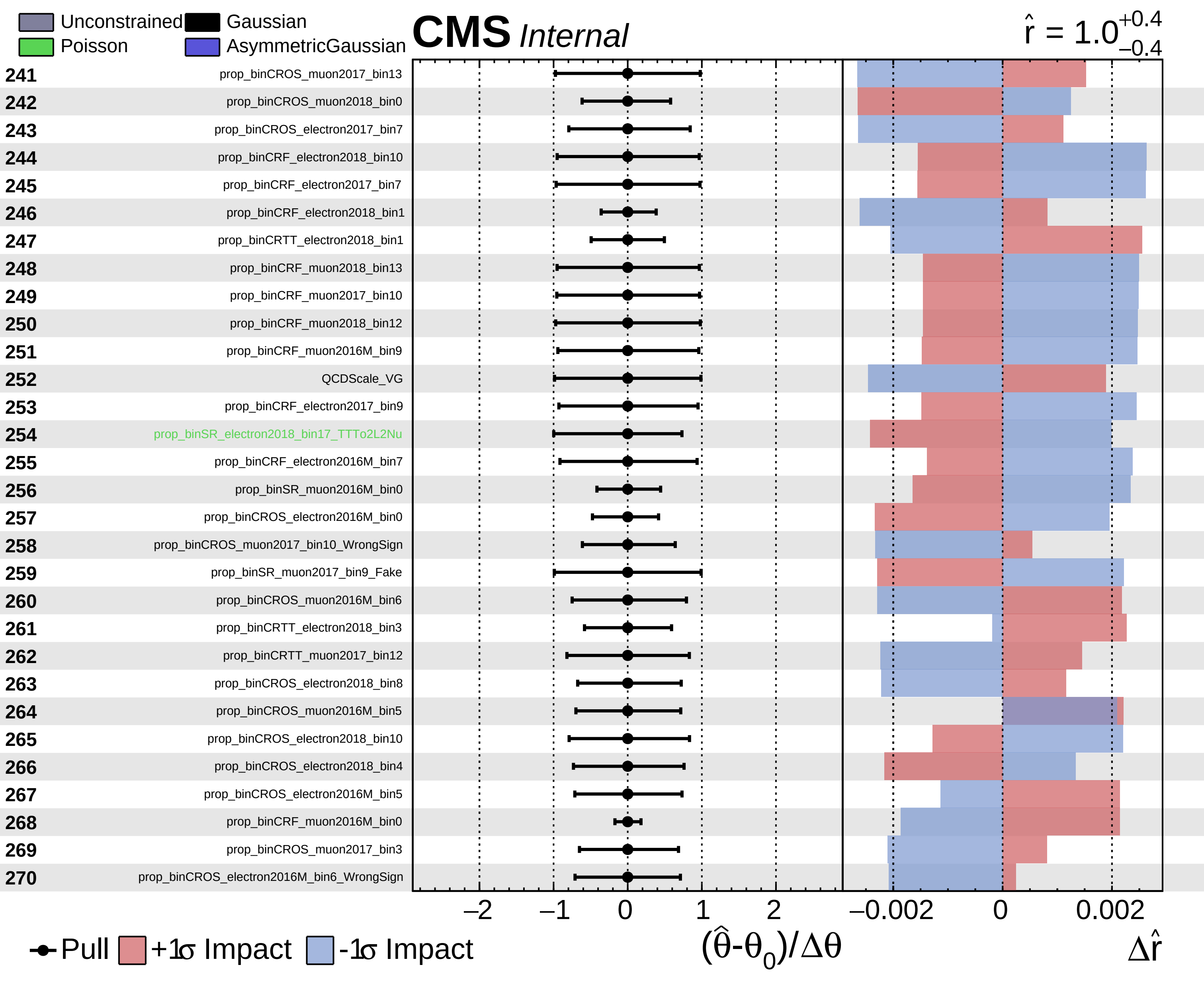


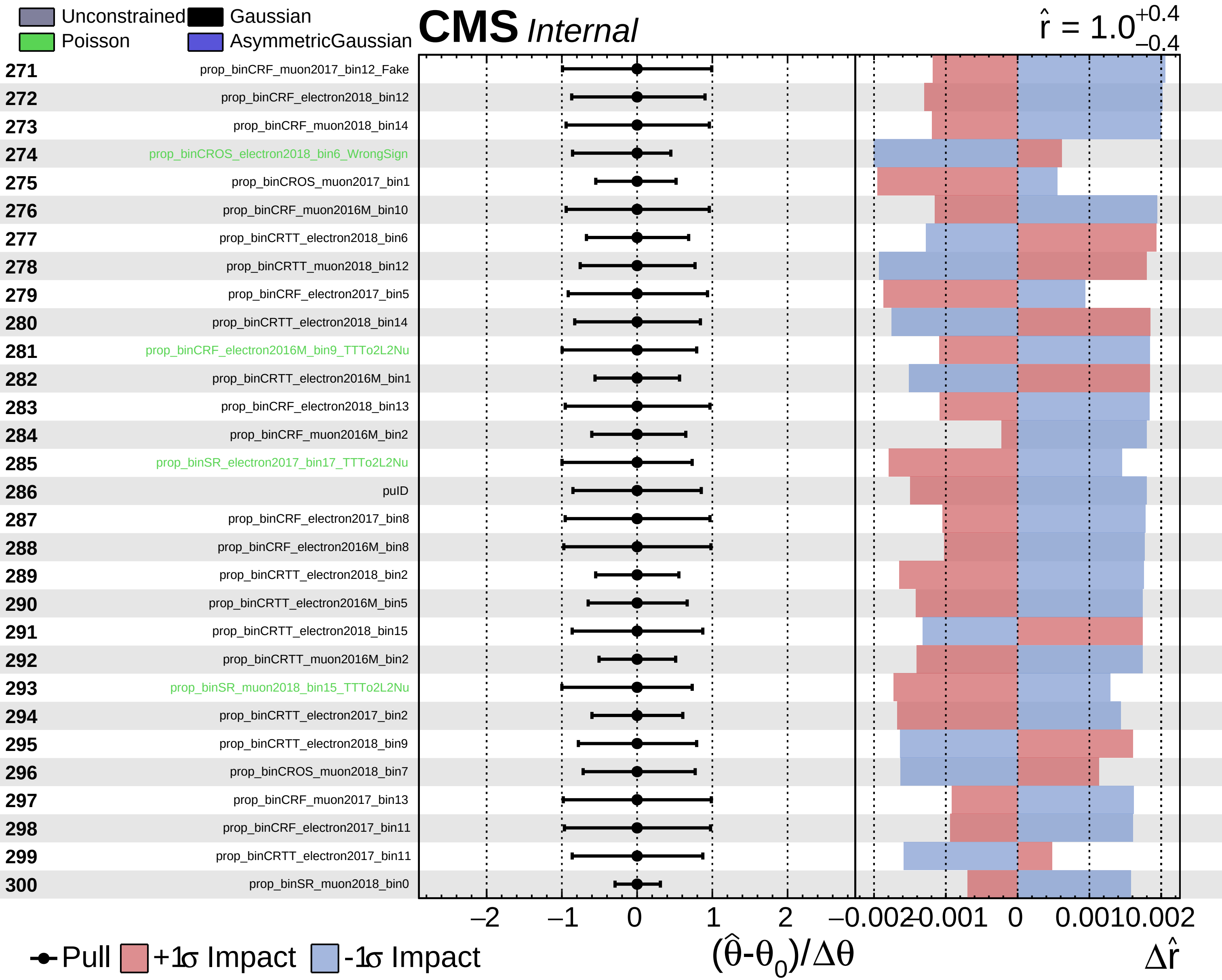
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.4}_{-0.4}$
151 prop_binCRF_muon2018_bin8
152 prop_binCRTT_muon2017_bin3
153 prop_binCROS_electron2018_bin7
154 prop_binCROS_muon2016M_bin0
155 prop_binCRF_electron2018_bin6
156 prop_binCRF_electron2017_bin2
157 TTest_muon_2018 $1.00^{+0.05}_{-0.05}$
158 prop_binCRF_electron2017_bin0
159 TTest_electron_2016M $1.00^{+0.06}_{-0.06}$
160 VBS_2016M
161 prop_binSR_electron2016M_bin3
162 TES
163 prop_binCRF_electron2017_bin3
164 prop_binCRF_muon2016M_bin4
165 prop_binCRF_electron2018_bin5
166 prop_binCROS_muon2018_bin2
167 prop_binSR_muon2016M_bin16_WrongSign
168 prop_binCRTT_muon2017_bin2
169 prop_binCROS_muon2018_bin8
170 prop_binCRTT_muon2018_bin6
171 prop_binCRF_electron2017_bin6
172 prop_binCRTT_muon2016M_bin1
173 prop_binSR_muon2016M_bin17
174 prop_binCRF_muon2018_bin9
175 prop_binCROS_muon2016M_bin9
176 prop_binCROS_electron2016M_bin7
177 prop_binCRTT_electron2018_bin0
178 prop_binCRF_muon2017_bin6
179 prop_binCROS_muon2017_bin9_WrongSign
180 QCDScale_TTdilep
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



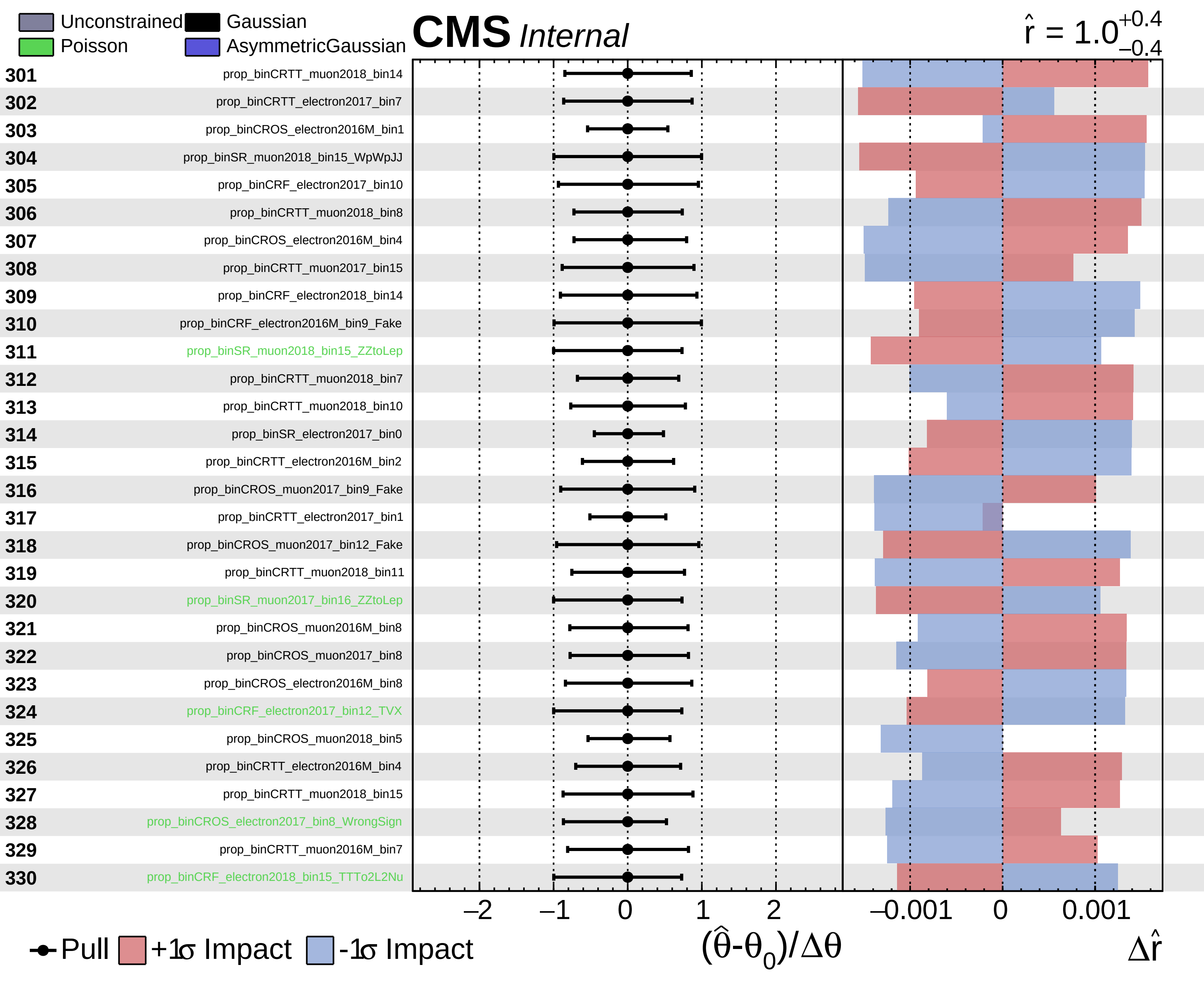
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.4}_{-0.4}$
181 prop_binSR_muon2016M_bin12_WrongSign
182 prop_binCROS_muon2017_bin11
183 TTest_muon_2016M
$1.00^{+0.06}_{-0.06}$
184 prop_binCROS_muon2016M_bin3
185 prop_binCRF_electron2018_bin7
186 prop_binCROS_muon2017_bin0
187 prop_binCROS_muon2017_bin12_WrongSign
188 prop_binCRF_electron2016M_bin2
189 prop_binCRF_electron2016M_bin10_VG
190 prop_binCRF_electron2017_bin4
191 prop_binCRF_muon2018_bin10
192 prop_binCRF_muon2017_bin7
193 prop_binCRF_electron2016M_bin0
194 prop_binCRF_muon2017_bin12_WZ
195 prop_binCRF_electron2018_bin2
196 prop_binCRF_muon2018_bin11
197 prop_binSR_electron2017_bin17_Triboson
198 prop_binCRF_electron2016M_bin3
199 prop_binCRTT_muon2018_bin0
200 prop_binCRTT_muon2018_bin1
201 prop_binCRF_muon2017_bin8
202 prop_binSR_electron2016M_bin2_Fake
203 prop_binCROS_electron2016M_bin15_WrongSign
204 prop_binCRF_muon2018_bin1
205 prop_binCROS_muon2017_bin5
206 prop_binCRF_muon2016M_bin7
207 prop_binCROS_muon2018_bin10
208 prop_binCRF_muon2016M_bin6
209 prop_binCRTT_muon2016M_bin3
210 prop_binCRF_electron2016M_bin4
−2 −1 0 1 2
−0.004 −0.002 0 0.002 0.004
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

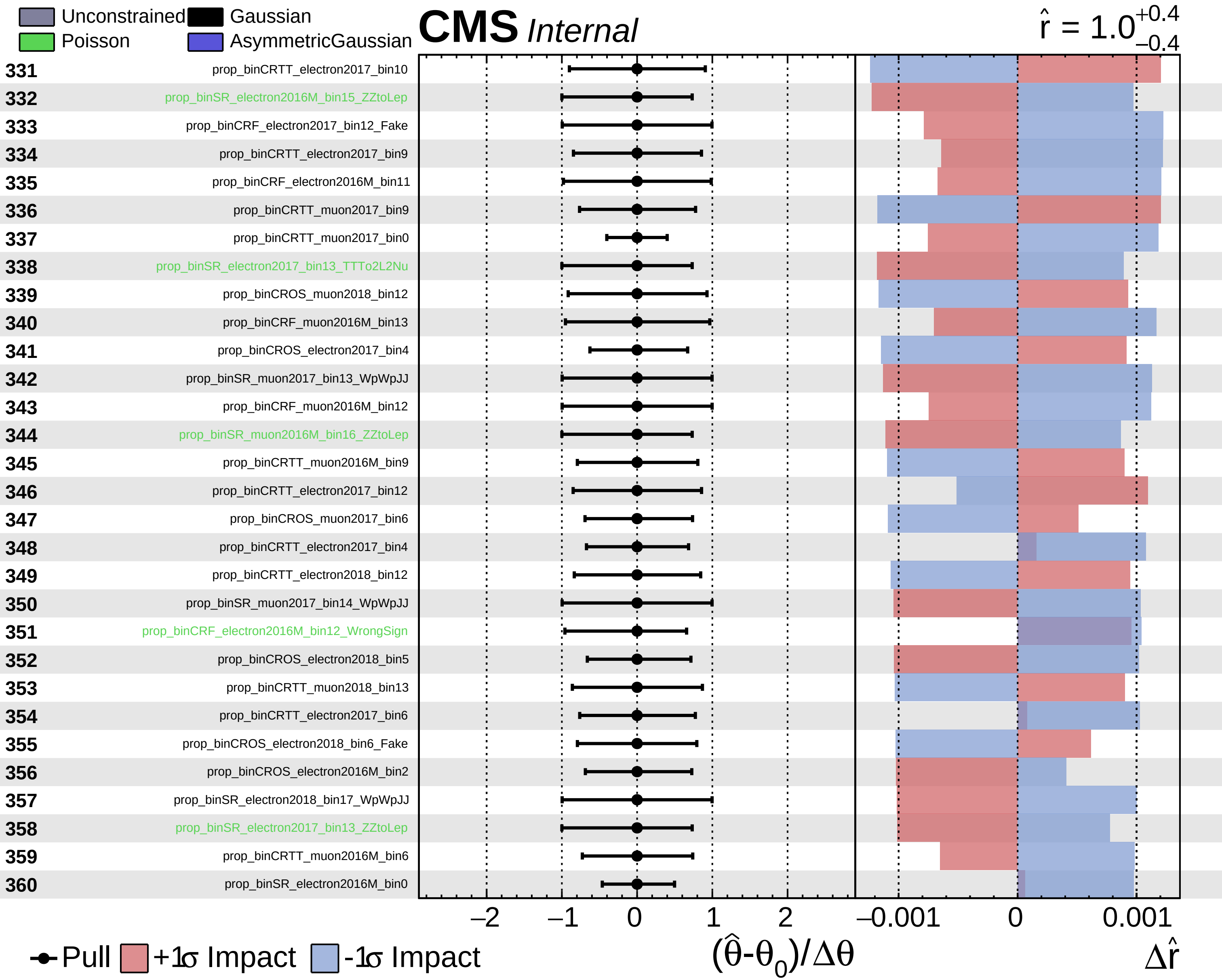

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.4}_{-0.4}$
211 prop_binCRTT_muon2018_bin2
212 prop_binCRF_electron2018_bin3
213 prop_binSR_electron2017_bin15_TTTo2L2Nu
214 prop_binSR_muon2017_bin0
215 prop_binSR_electron2018_bin16
216 prop_binCRTT_electron2017_bin3
217 prop_binCRF_electron2018_bin8
218 prop_binCRTT_muon2017_bin5
219 QCDScale_Triboson
220 prop_binCROS_muon2016M_bin2
221 prop_binSR_electron2016M_bin15_WZ
222 prop_binCRF_electron2016M_bin5
223 prop_binSR_muon2017_bin14_WrongSign
224 prop_binCRF_electron2018_bin4
225 prop_binCRTT_electron2018_bin8
226 prop_binCRTT_muon2017_bin1
227 jer_2017
228 prop_binCRF_muon2016M_bin1
229 prop_binSR_electron2016M_bin2_WrongSign
230 prop_binCROS_electron2017_bin2
231 prop_binCRF_electron2018_bin9
232 prop_binCROS_electron2017_bin0
233 prop_binCRF_muon2016M_bin8
234 prop_binSR_electron2016M_bin16_WZ
235 prop_binCRF_electron2016M_bin6
236 prop_binCRF_muon2017_bin0
237 prop_binSR_muon2016M_bin15_TTTo2L2Nu
238 prop_binCRF_electron2017_bin1
239 prop_binCROS_electron2017_bin9_WrongSign
240 prop_binCROS_muon2018_bin6
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.4}_{-0.4}$
241 prop_binCROS_muon2017_bin13
242 prop_binCROS_muon2018_bin0
243 prop_binCROS_electron2017_bin7
244 prop_binCRF_electron2018_bin10
245 prop_binCRF_electron2017_bin7
246 prop_binCRF_electron2018_bin1
247 prop_binCRTT_electron2018_bin1
248 prop_binCRF_muon2018_bin13
249 prop_binCRF_muon2017_bin10
250 prop_binCRF_muon2018_bin12
251 prop_binCRF_muon2016M_bin9
252 QCDScale_VG
253 prop_binCRF_electron2017_bin9
254 prop_binSR_electron2018_bin17_TTTo2L2Nu
255 prop_binCRF_electron2016M_bin7
256 prop_binSR_muon2016M_bin0
257 prop_binCROS_electron2016M_bin0
258 prop_binCROS_muon2017_bin10_WrongSign
259 prop_binSR_muon2017_bin9_Fake
260 prop_binCROS_muon2016M_bin6
261 prop_binCRTT_electron2018_bin3
262 prop_binCRTT_muon2017_bin12
263 prop_binCROS_electron2018_bin8
264 prop_binCROS_muon2016M_bin5
265 prop_binCROS_electron2018_bin10
266 prop_binCROS_electron2018_bin4
267 prop_binCROS_electron2016M_bin5
268 prop_binCRF_muon2016M_bin0
269 prop_binCROS_muon2017_bin3
270 prop_binCROS_electron2016M_bin6_WrongSign
−2 −1 0 1 2
−0.002 0 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

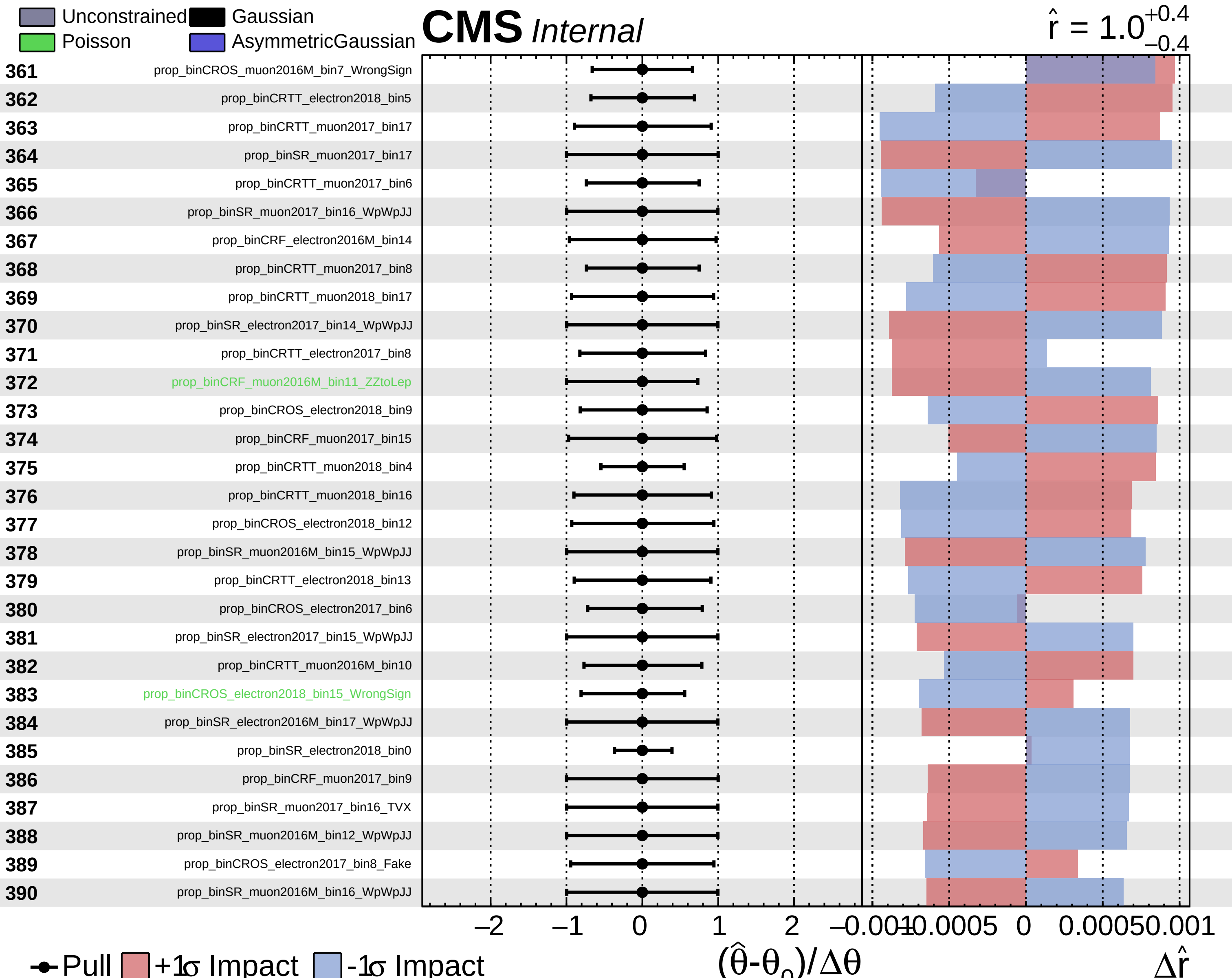


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.4}_{-0.4}$
271 prop_binCRF_muon2017_bin12_Fake
272 prop_binCRF_electron2018_bin12
273 prop_binCRF_muon2018_bin14
274 prop_binCROS_electron2018_bin6_WrongSign
275 prop_binCROS_muon2017_bin1
276 prop_binCRF_muon2016M_bin10
277 prop_binCRTT_electron2018_bin6
278 prop_binCRTT_muon2018_bin12
279 prop_binCRF_electron2017_bin5
280 prop_binCRTT_electron2018_bin14
281 prop_binCRF_electron2016M_bin9_TTTo2L2Nu
282 prop_binCRTT_electron2016M_bin1
283 prop_binCRF_electron2018_bin13
284 prop_binCRF_muon2016M_bin2
285 prop_binSR_electron2017_bin17_TTTo2L2Nu
286 puID
287 prop_binCRF_electron2017_bin8
288 prop_binCRF_electron2016M_bin8
289 prop_binCRTT_electron2018_bin2
290 prop_binCRTT_electron2016M_bin5
291 prop_binCRTT_electron2018_bin15
292 prop_binCRTT_muon2016M_bin2
293 prop_binSR_muon2018_bin15_TTTo2L2Nu
294 prop_binCRTT_electron2017_bin2
295 prop_binCRTT_electron2018_bin9
296 prop_binCROS_muon2018_bin7
297 prop_binCRF_muon2017_bin13
298 prop_binCRF_electron2017_bin11
299 prop_binCRTT_electron2017_bin11
300 prop_binSR_muon2018_bin0
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

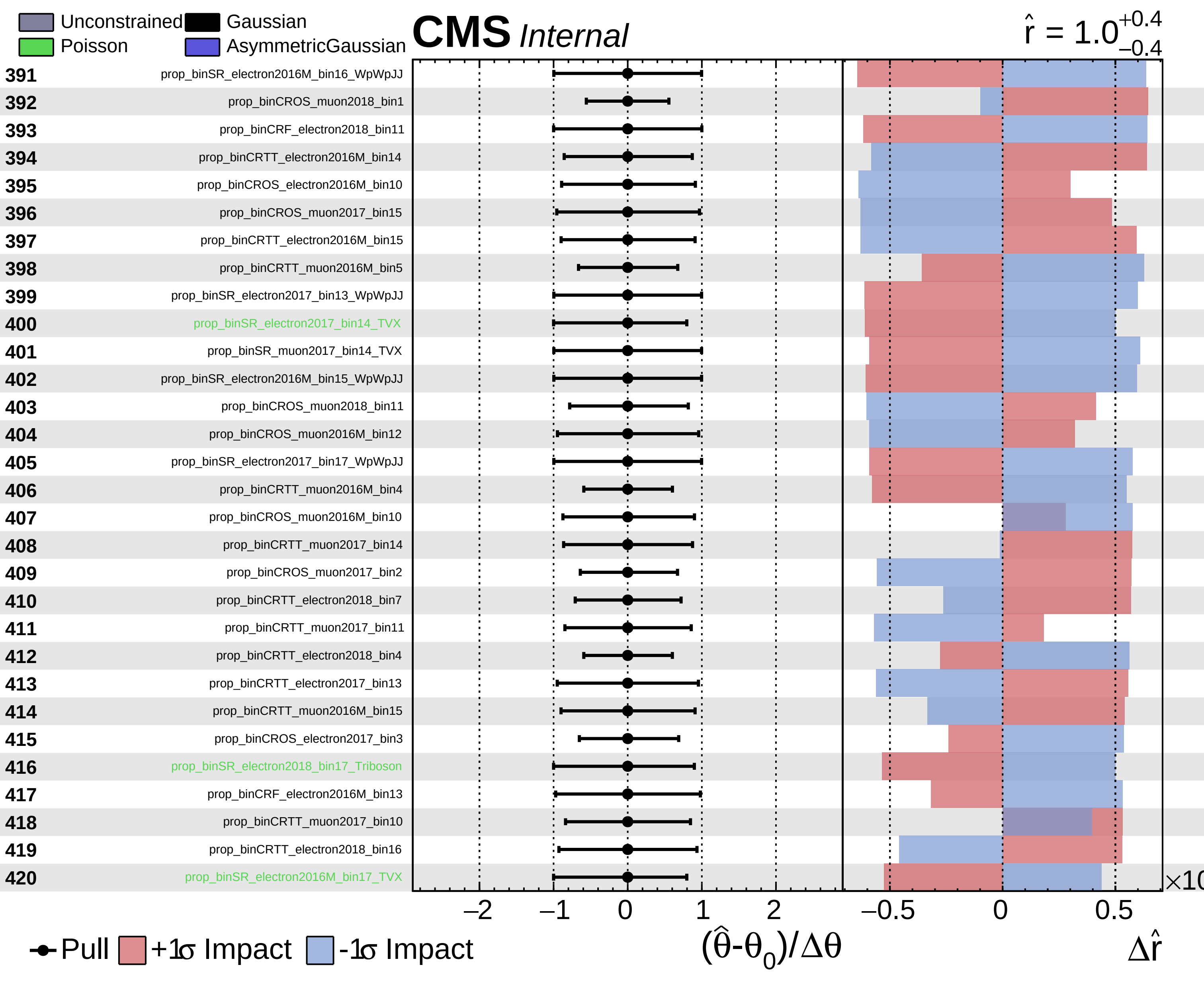

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 1.0 +0.4 −0.4
301 prop_binCRTT_muon2018_bin14
302 prop_binCRTT_electron2017_bin7
303 prop_binCROS_electron2016M_bin1
304 prop_binSR_muon2018_bin15_WpWpJJ
305 prop_binCRF_electron2017_bin10
306 prop_binCRTT_muon2018_bin8
307 prop_binCROS_electron2016M_bin4
308 prop_binCRTT_muon2017_bin15
309 prop_binCRF_electron2018_bin14
310 prop_binCRF_electron2016M_bin9_Fake
311 prop_binSR_muon2018_bin15_ZZtoLep
312 prop_binCRTT_muon2018_bin7
313 prop_binCRTT_muon2018_bin10
314 prop_binSR_electron2017_bin0
315 prop_binCRTT_electron2016M_bin2
316 prop_binCROS_muon2017_bin9_Fake
317 prop_binCRTT_electron2017_bin1
318 prop_binCROS_muon2017_bin12_Fake
319 prop_binCRTT_muon2018_bin11
320 prop_binSR_muon2017_bin16_ZZtoLep
321 prop_binCROS_muon2016M_bin8
322 prop_binCROS_muon2017_bin8
323 prop_binCROS_electron2016M_bin8
324 prop_binCRF_electron2017_bin12_TVX
325 prop_binCROS_muon2018_bin5
326 prop_binCRTT_electron2016M_bin4
327 prop_binCRTT_muon2018_bin15
328 prop_binCROS_electron2017_bin8_WrongSign
329 prop_binCRTT_muon2016M_bin7
330 prop_binCRF_electron2018_bin15_TTTo2L2Nu
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂



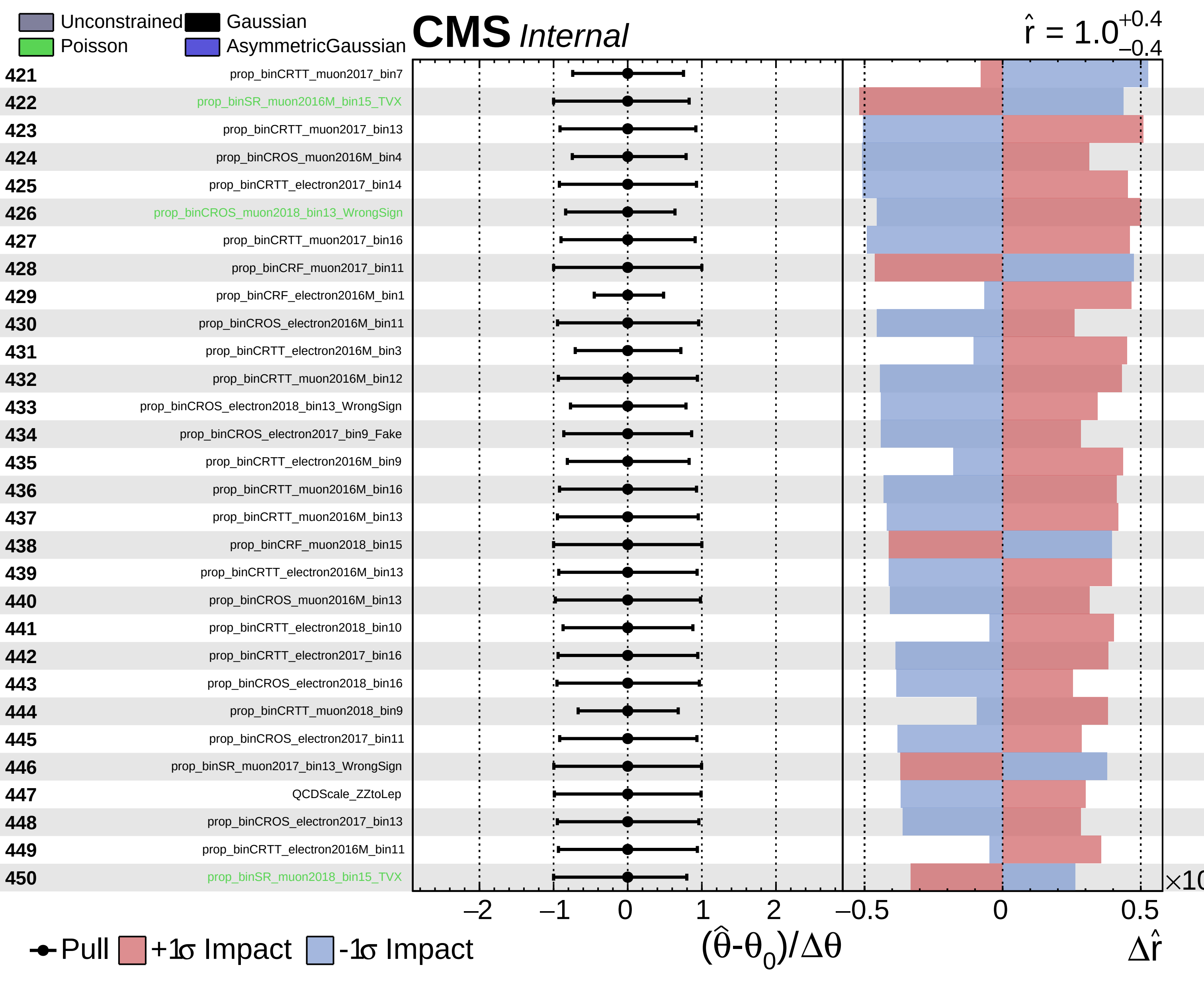
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.4}_{-0.4}$
331 prop_binCRTT_electron2017_bin10
332 prop_binSR_electron2016M_bin15_ZZtoLep
333 prop_binCRF_electron2017_bin12_Fake
334 prop_binCRTT_electron2017_bin9
335 prop_binCRF_electron2016M_bin11
336 prop_binCRTT_muon2017_bin9
337 prop_binCRTT_muon2017_bin0
338 prop_binSR_electron2017_bin13_TTTo2L2Nu
339 prop_binCROS_muon2018_bin12
340 prop_binCRF_muon2016M_bin13
341 prop_binCROS_electron2017_bin4
342 prop_binSR_muon2017_bin13_WpWpJJ
343 prop_binCRF_muon2016M_bin12
344 prop_binSR_muon2016M_bin16_ZZtoLep
345 prop_binCRTT_muon2016M_bin9
346 prop_binCRTT_electron2017_bin12
347 prop_binCROS_muon2017_bin6
348 prop_binCRTT_electron2017_bin4
349 prop_binCRTT_electron2018_bin12
350 prop_binSR_muon2017_bin14_WpWpJJ
351 prop_binCRF_electron2016M_bin12_WrongSign
352 prop_binCROS_electron2018_bin5
353 prop_binCRTT_muon2018_bin13
354 prop_binCRTT_electron2017_bin6
355 prop_binCROS_electron2018_bin6_Fake
356 prop_binCROS_electron2016M_bin2
357 prop_binSR_electron2018_bin17_WpWpJJ
358 prop_binSR_electron2017_bin13_ZZtoLep
359 prop_binCRTT_muon2016M_bin6
360 prop_binSR_electron2016M_bin0
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



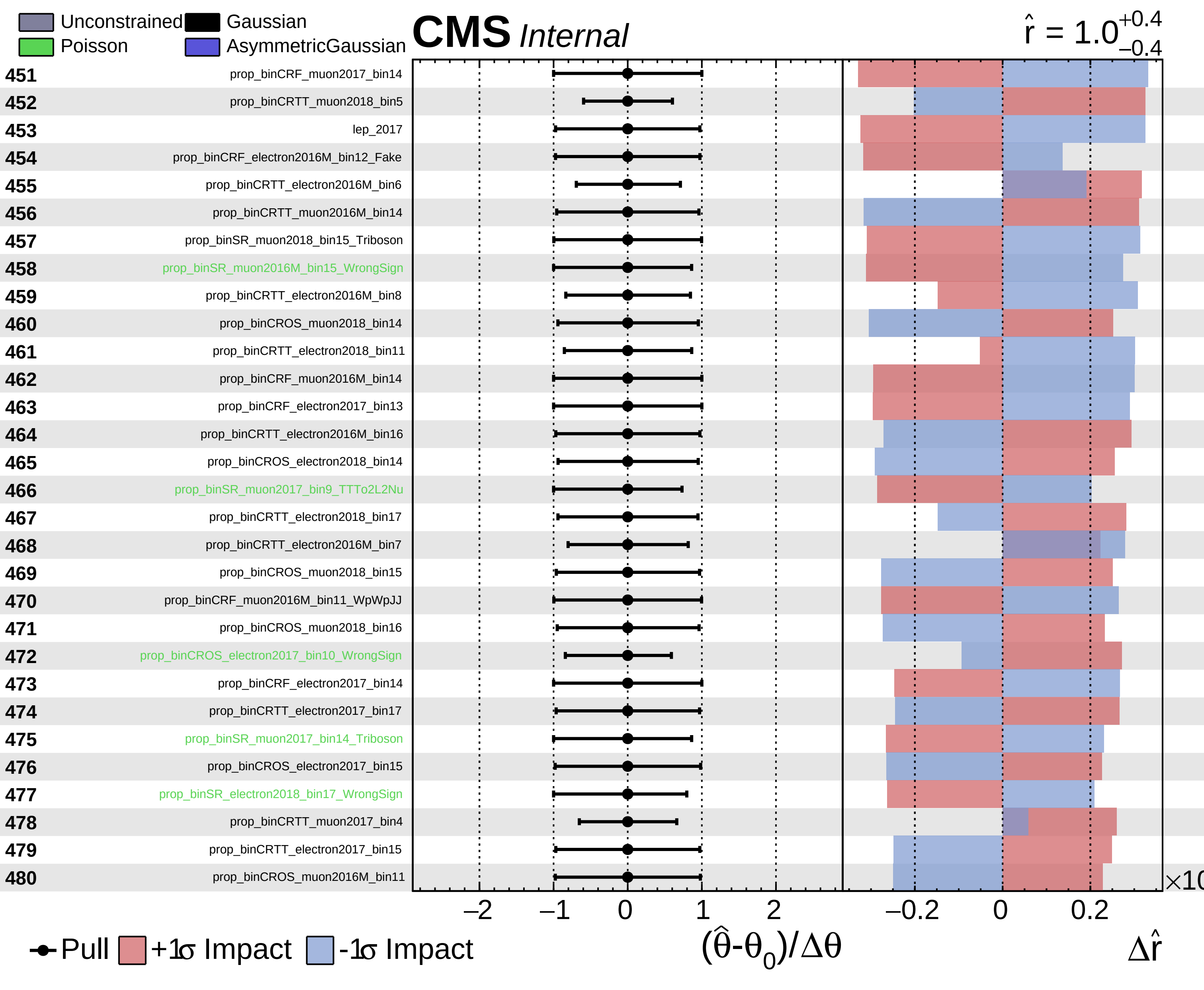
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.4}_{-0.4}$
361 prop_binCROS_muon2016M_bin7_WrongSign
362 prop_binCRTT_electron2018_bin5
363 prop_binCRTT_muon2017_bin17
364 prop_binSR_muon2017_bin17
365 prop_binCRTT_muon2017_bin6
366 prop_binSR_muon2017_bin16_WpWpJJ
367 prop_binCRF_electron2016M_bin14
368 prop_binCRTT_muon2017_bin8
369 prop_binCRTT_muon2018_bin17
370 prop_binSR_electron2017_bin14_WpWpJJ
371 prop_binCRTT_electron2017_bin8
372 prop_binCRF_muon2016M_bin11_ZZtoLep
373 prop_binCROS_electron2018_bin9
374 prop_binCRF_muon2017_bin15
375 prop_binCRTT_muon2018_bin4
376 prop_binCRTT_muon2018_bin16
377 prop_binCROS_electron2018_bin12
378 prop_binSR_muon2016M_bin15_WpWpJJ
379 prop_binCRTT_electron2018_bin13
380 prop_binCROS_electron2017_bin6
381 prop_binSR_electron2017_bin15_WpWpJJ
382 prop_binCRTT_muon2016M_bin10
383 prop_binCROS_electron2018_bin15_WrongSign
384 prop_binSR_electron2016M_bin17_WpWpJJ
385 prop_binSR_electron2018_bin0
386 prop_binCRF_muon2017_bin9
387 prop_binSR_muon2017_bin16_TVX
388 prop_binSR_muon2016M_bin12_WpWpJJ
389 prop_binCROS_electron2017_bin8_Fake
390 prop_binSR_muon2016M_bin16_WpWpJJ
−2
−1
0
1
2
−0.001
−0.0005
0
0.0005
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



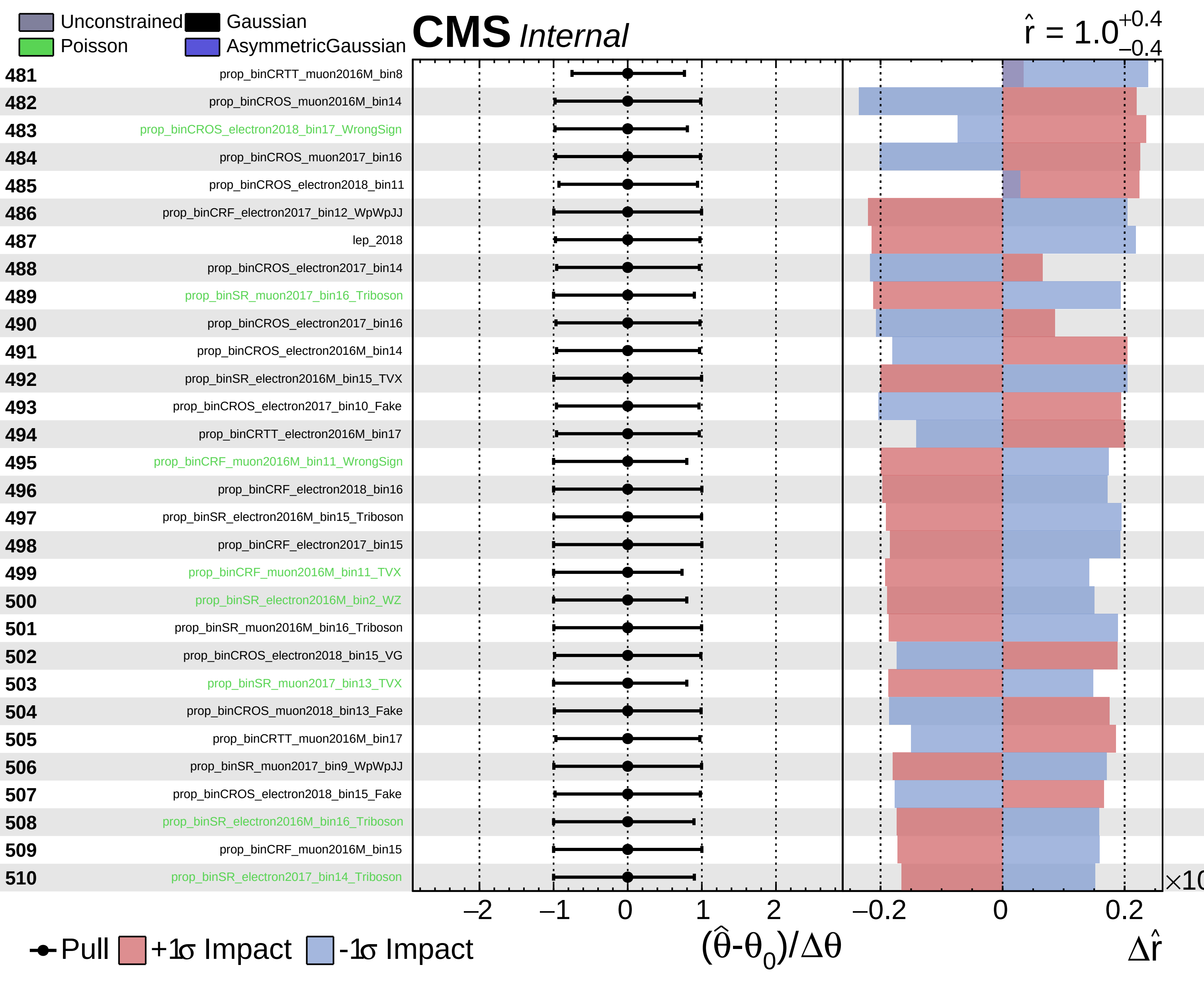
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.4}_{-0.4}$
391 prop_binSR_electron2016M_bin16_WpWpJJ
392 prop_binCROS_muon2018_bin1
393 prop_binCRF_electron2018_bin11
394 prop_binCRTT_electron2016M_bin14
395 prop_binCROS_electron2016M_bin10
396 prop_binCROS_muon2017_bin15
397 prop_binCRTT_electron2016M_bin15
398 prop_binCRTT_muon2016M_bin5
399 prop_binSR_electron2017_bin13_WpWpJJ
400 prop_binSR_electron2017_bin14_TVX
401 prop_binSR_muon2017_bin14_TVX
402 prop_binSR_electron2016M_bin15_WpWpJJ
403 prop_binCROS_muon2018_bin11
404 prop_binCROS_muon2016M_bin12
405 prop_binSR_electron2017_bin17_WpWpJJ
406 prop_binCRTT_muon2016M_bin4
407 prop_binCROS_muon2016M_bin10
408 prop_binCRTT_muon2017_bin14
409 prop_binCROS_muon2017_bin2
410 prop_binCRTT_electron2018_bin7
411 prop_binCRTT_muon2017_bin11
412 prop_binCRTT_electron2018_bin4
413 prop_binCRTT_electron2017_bin13
414 prop_binCRTT_muon2016M_bin15
415 prop_binCROS_electron2017_bin3
416 prop_binSR_electron2018_bin17_Triboson
417 prop_binCRF_electron2016M_bin13
418 prop_binCRTT_muon2017_bin10
419 prop_binCRTT_electron2018_bin16
420 prop_binSR_electron2016M_bin17_TVX
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



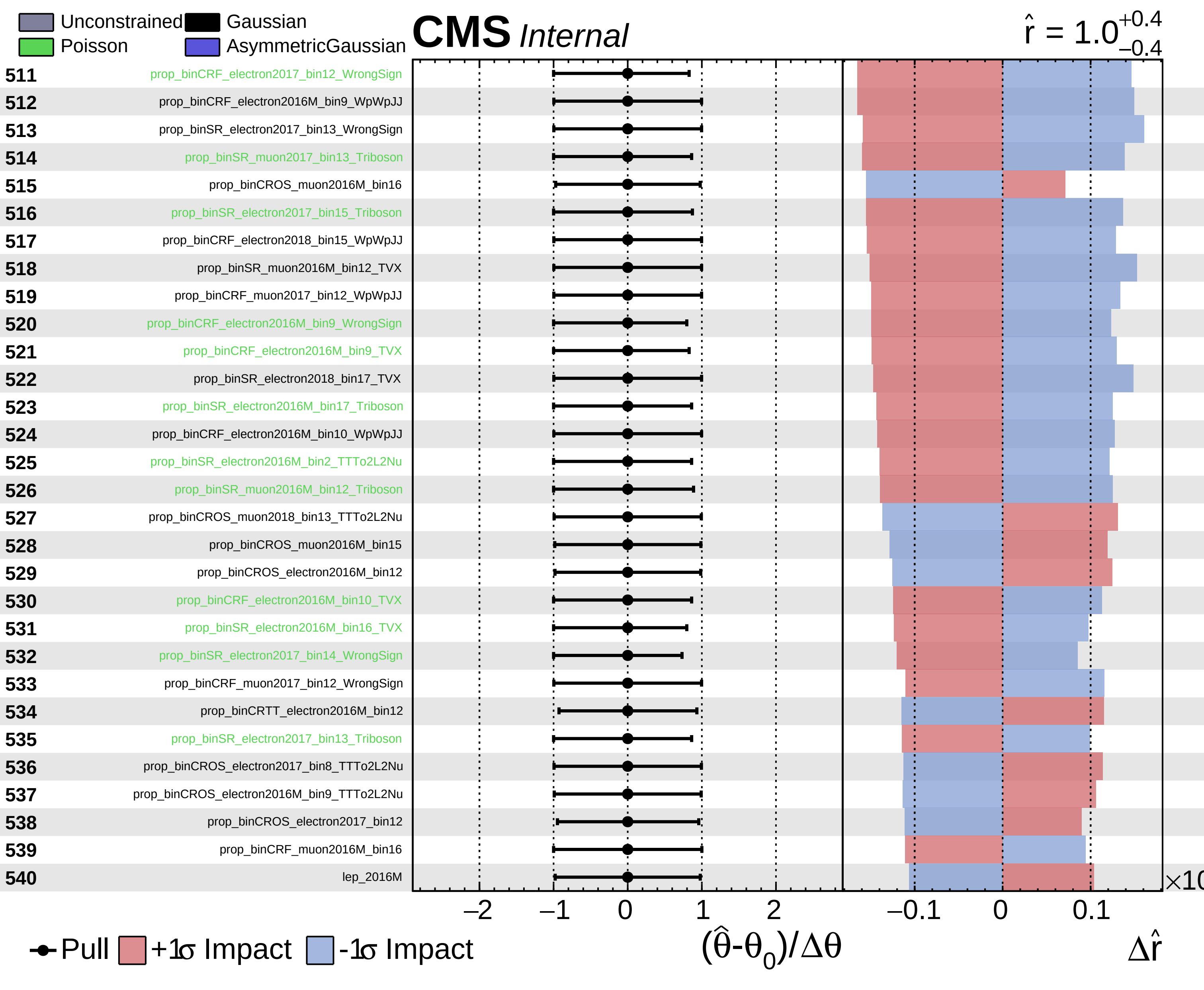
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.4}_{-0.4}$
421 prop_binCRTT_muon2017_bin7
422 prop_binSR_muon2016M_bin15_TVX
423 prop_binCRTT_muon2017_bin13
424 prop_binCROS_muon2016M_bin4
425 prop_binCRTT_electron2017_bin14
426 prop_binCROS_muon2018_bin13_WrongSign
427 prop_binCRTT_muon2017_bin16
428 prop_binCRF_muon2017_bin11
429 prop_binCRF_electron2016M_bin1
430 prop_binCROS_electron2016M_bin11
431 prop_binCRTT_electron2016M_bin3
432 prop_binCRTT_muon2016M_bin12
433 prop_binCROS_electron2018_bin13_WrongSign
434 prop_binCROS_electron2017_bin9_Fake
435 prop_binCRTT_electron2016M_bin9
436 prop_binCRTT_muon2016M_bin16
437 prop_binCRTT_muon2016M_bin13
438 prop_binCRF_muon2018_bin15
439 prop_binCRTT_electron2016M_bin13
440 prop_binCROS_muon2016M_bin13
441 prop_binCRTT_electron2018_bin10
442 prop_binCRTT_electron2017_bin16
443 prop_binCROS_electron2018_bin16
444 prop_binCRTT_muon2018_bin9
445 prop_binCROS_electron2017_bin11
446 prop_binSR_muon2017_bin13_WrongSign
447 QCDScale_ZZtoLep
448 prop_binCROS_electron2017_bin13
449 prop_binCRTT_electron2016M_bin11
450 prop_binSR_muon2018_bin15_TVX
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



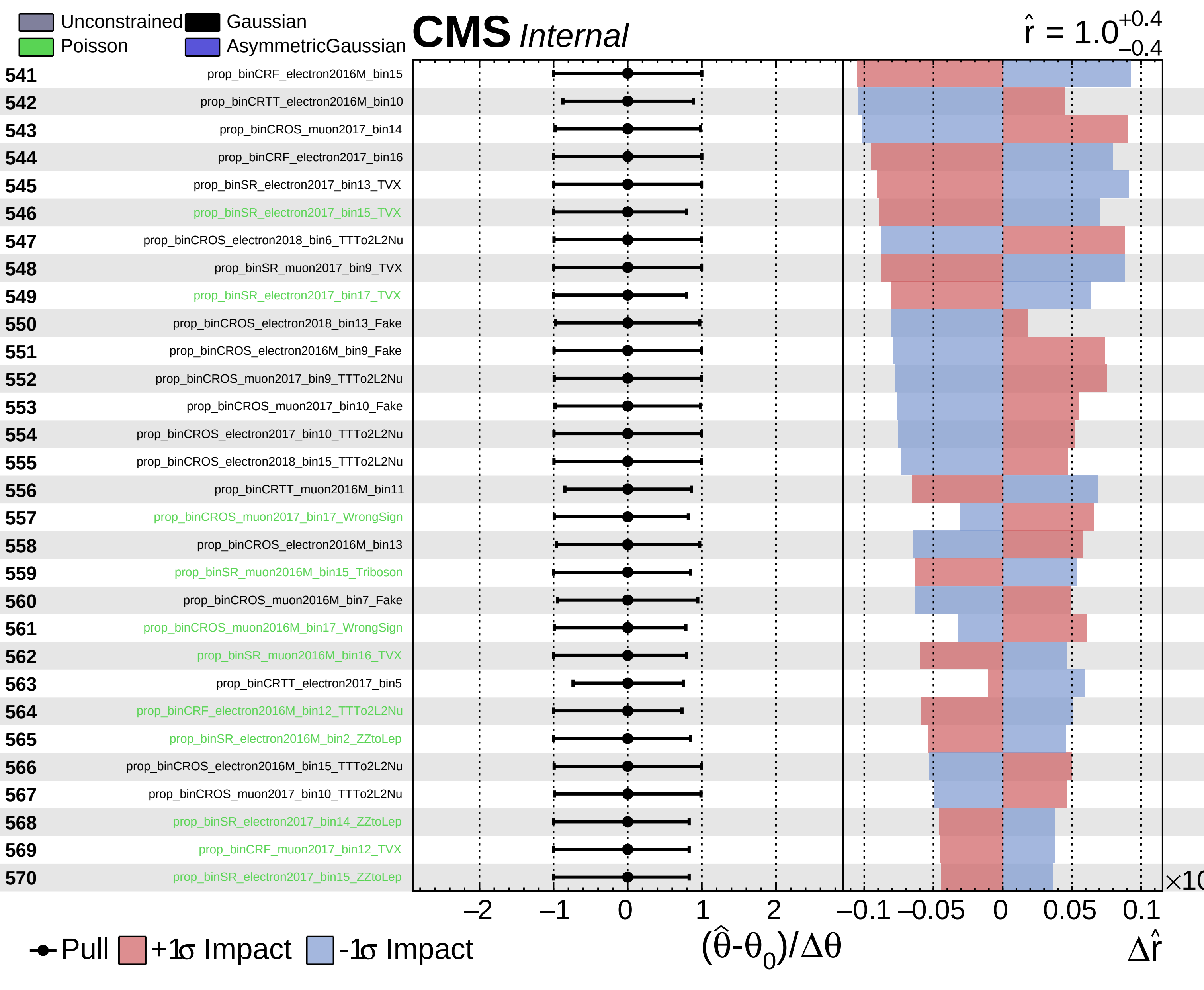
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.4}_{-0.4}$
451 prop_binCRF_muon2017_bin14
452 prop_binCRTT_muon2018_bin5
453 lep_2017
454 prop_binCRF_electron2016M_bin12_Fake
455 prop_binCRTT_electron2016M_bin6
456 prop_binCRTT_muon2016M_bin14
457 prop_binSR_muon2018_bin15_Triboson
458 prop_binSR_muon2016M_bin15_WrongSign
459 prop_binCRTT_electron2016M_bin8
460 prop_binCROS_muon2018_bin14
461 prop_binCRTT_electron2018_bin11
462 prop_binCRF_muon2016M_bin14
463 prop_binCRF_electron2017_bin13
464 prop_binCRTT_electron2016M_bin16
465 prop_binCROS_electron2018_bin14
466 prop_binSR_muon2017_bin9_TTTo2L2Nu
467 prop_binCRTT_electron2018_bin17
468 prop_binCRTT_electron2016M_bin7
469 prop_binCROS_muon2018_bin15
470 prop_binCRF_muon2016M_bin11_WpWpJJ
471 prop_binCROS_muon2018_bin16
472 prop_binCROS_electron2017_bin10_WrongSign
473 prop_binCRF_electron2017_bin14
474 prop_binCRTT_electron2017_bin17
475 prop_binSR_muon2017_bin14_Triboson
476 prop_binCROS_electron2017_bin15
477 prop_binSR_electron2018_bin17_WrongSign
478 prop_binCRTT_muon2017_bin4
479 prop_binCRTT_electron2017_bin15
480 prop_binCROS_muon2016M_bin11
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



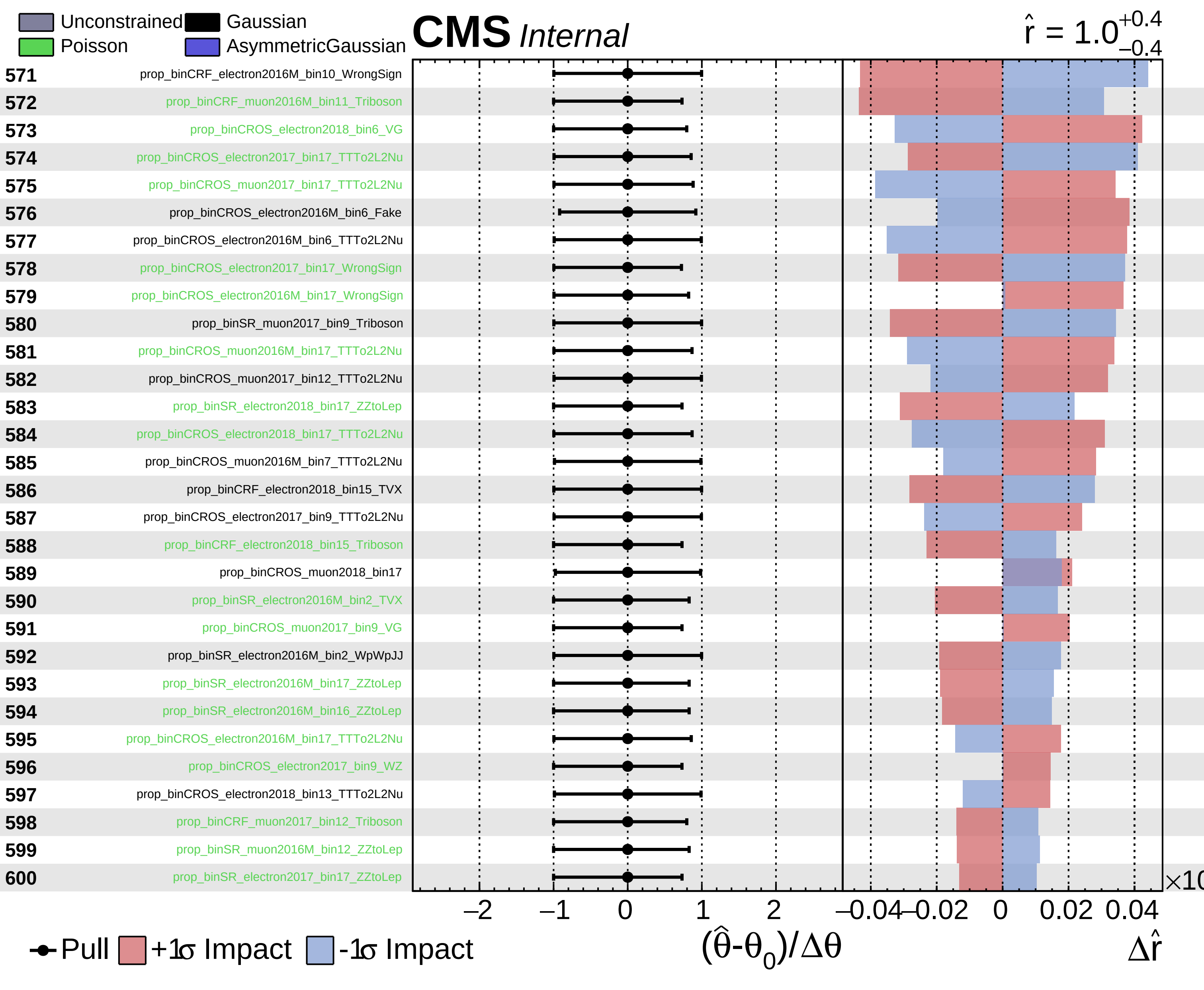
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.4}_{-0.4}$
481 prop_binCRTT_muon2016M_bin8
482 prop_binCROS_muon2016M_bin14
483 prop_binCROS_electron2018_bin17_WrongSign
484 prop_binCROS_muon2017_bin16
485 prop_binCROS_electron2018_bin11
486 prop_binCRF_electron2017_bin12_WpWpJJ
487 lep_2018
488 prop_binCROS_electron2017_bin14
489 prop_binSR_muon2017_bin16_Triboson
490 prop_binCROS_electron2017_bin16
491 prop_binCROS_electron2016M_bin14
492 prop_binSR_electron2016M_bin15_TVX
493 prop_binCROS_electron2017_bin10_Fake
494 prop_binCRTT_electron2016M_bin17
495 prop_binCRF_muon2016M_bin11_WrongSign
496 prop_binCRF_electron2018_bin16
497 prop_binSR_electron2016M_bin15_Triboson
498 prop_binCRF_electron2017_bin15
499 prop_binCRF_muon2016M_bin11_TVX
500 prop_binSR_electron2016M_bin2_WZ
501 prop_binSR_muon2016M_bin16_Triboson
502 prop_binCROS_electron2018_bin15_VG
503 prop_binSR_muon2017_bin13_TVX
504 prop_binCROS_muon2018_bin13_Fake
505 prop_binCRTT_muon2016M_bin17
506 prop_binSR_muon2017_bin9_WpWpJJ
507 prop_binCROS_electron2018_bin15_Fake
508 prop_binSR_electron2016M_bin16_Triboson
509 prop_binCRF_muon2016M_bin15
510 prop_binSR_electron2017_bin14_Triboson
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



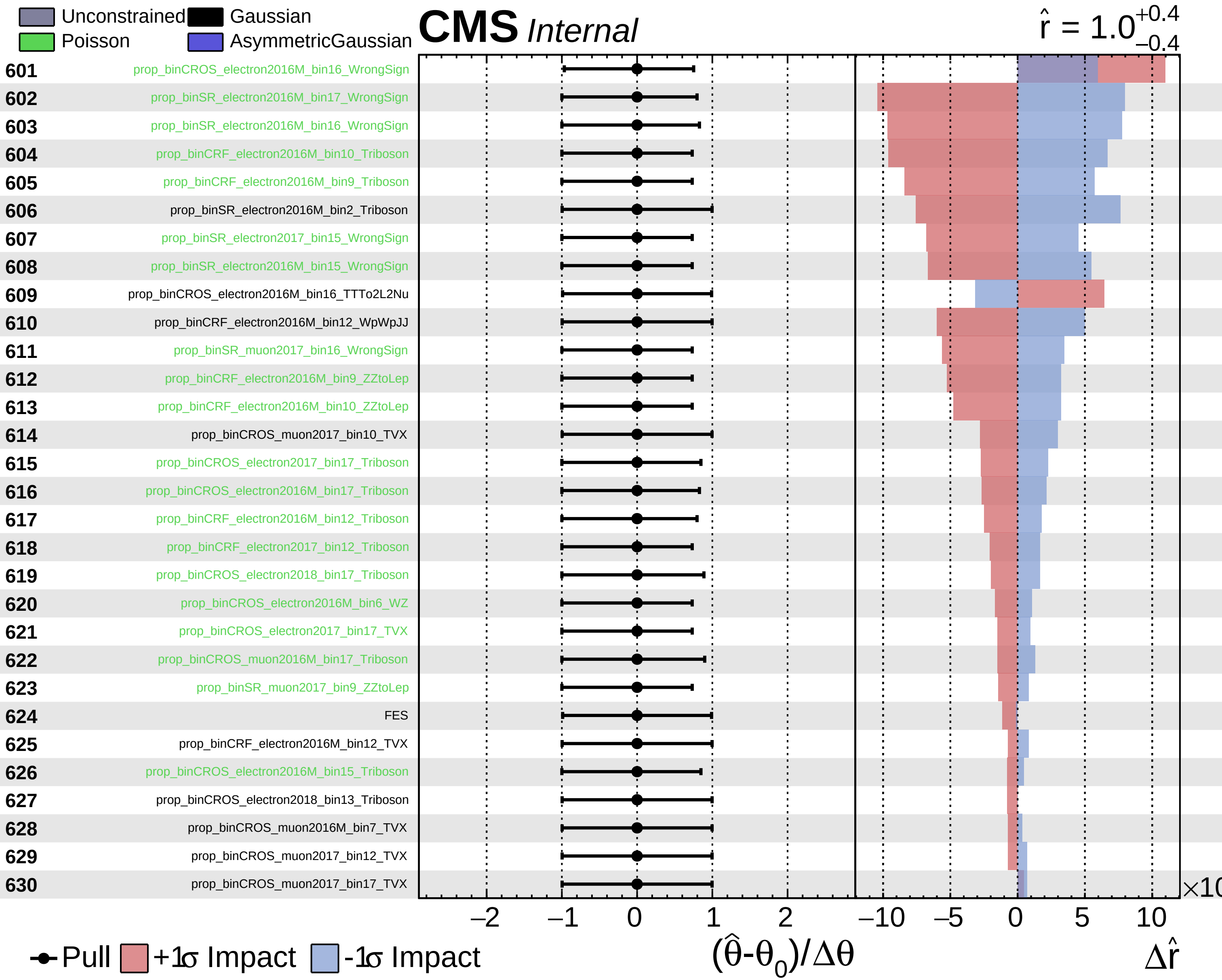
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 1.0 +0.4 −0.4
511 prop_binCRF_electron2017_bin12_WrongSign
512 prop_binCRF_electron2016M_bin9_WpWpJJ
513 prop_binSR_electron2017_bin13_WrongSign
514 prop_binSR_muon2017_bin13_Triboson
515 prop_binCROS_muon2016M_bin16
516 prop_binSR_electron2017_bin15_Triboson
517 prop_binCRF_electron2018_bin15_WpWpJJ
518 prop_binSR_muon2016M_bin12_TVX
519 prop_binCRF_muon2017_bin12_WpWpJJ
520 prop_binCRF_electron2016M_bin9_WrongSign
521 prop_binCRF_electron2016M_bin9_TVX
522 prop_binSR_electron2018_bin17_TVX
523 prop_binSR_electron2016M_bin17_Triboson
524 prop_binCRF_electron2016M_bin10_WpWpJJ
525 prop_binSR_electron2016M_bin2_TTTo2L2Nu
526 prop_binSR_muon2016M_bin12_Triboson
527 prop_binCROS_muon2018_bin13_TTTo2L2Nu
528 prop_binCROS_muon2016M_bin15
529 prop_binCROS_electron2016M_bin12
530 prop_binCRF_electron2016M_bin10_TVX
531 prop_binSR_electron2016M_bin16_TVX
532 prop_binSR_electron2017_bin14_WrongSign
533 prop_binCRF_muon2017_bin12_WrongSign
534 prop_binCRTT_electron2016M_bin12
535 prop_binSR_electron2017_bin13_Triboson
536 prop_binCROS_electron2017_bin8_TTTo2L2Nu
537 prop_binCROS_electron2016M_bin9_TTTo2L2Nu
538 prop_binCROS_electron2017_bin12
539 prop_binCRF_muon2016M_bin16
540 lep_2016M
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂

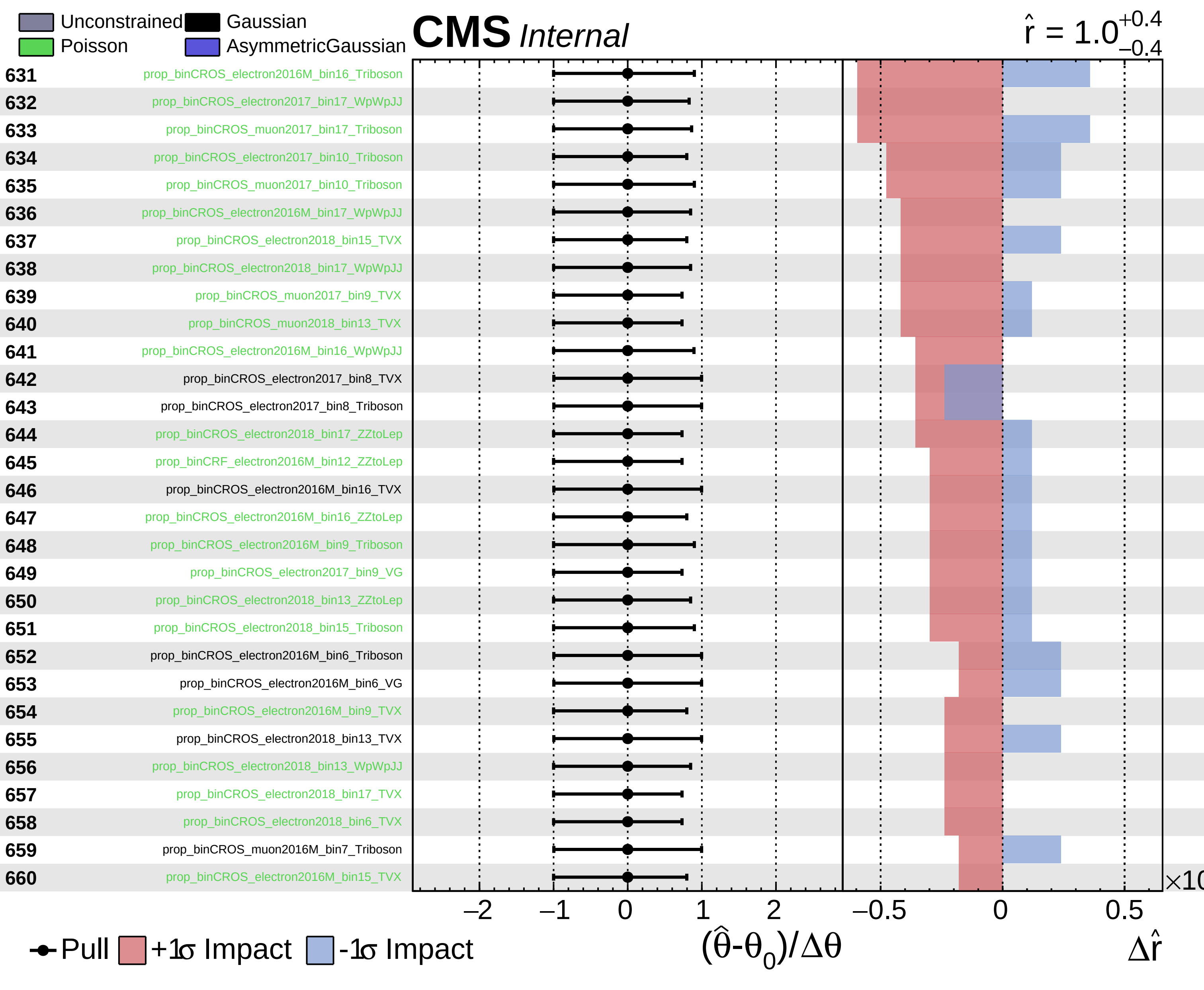


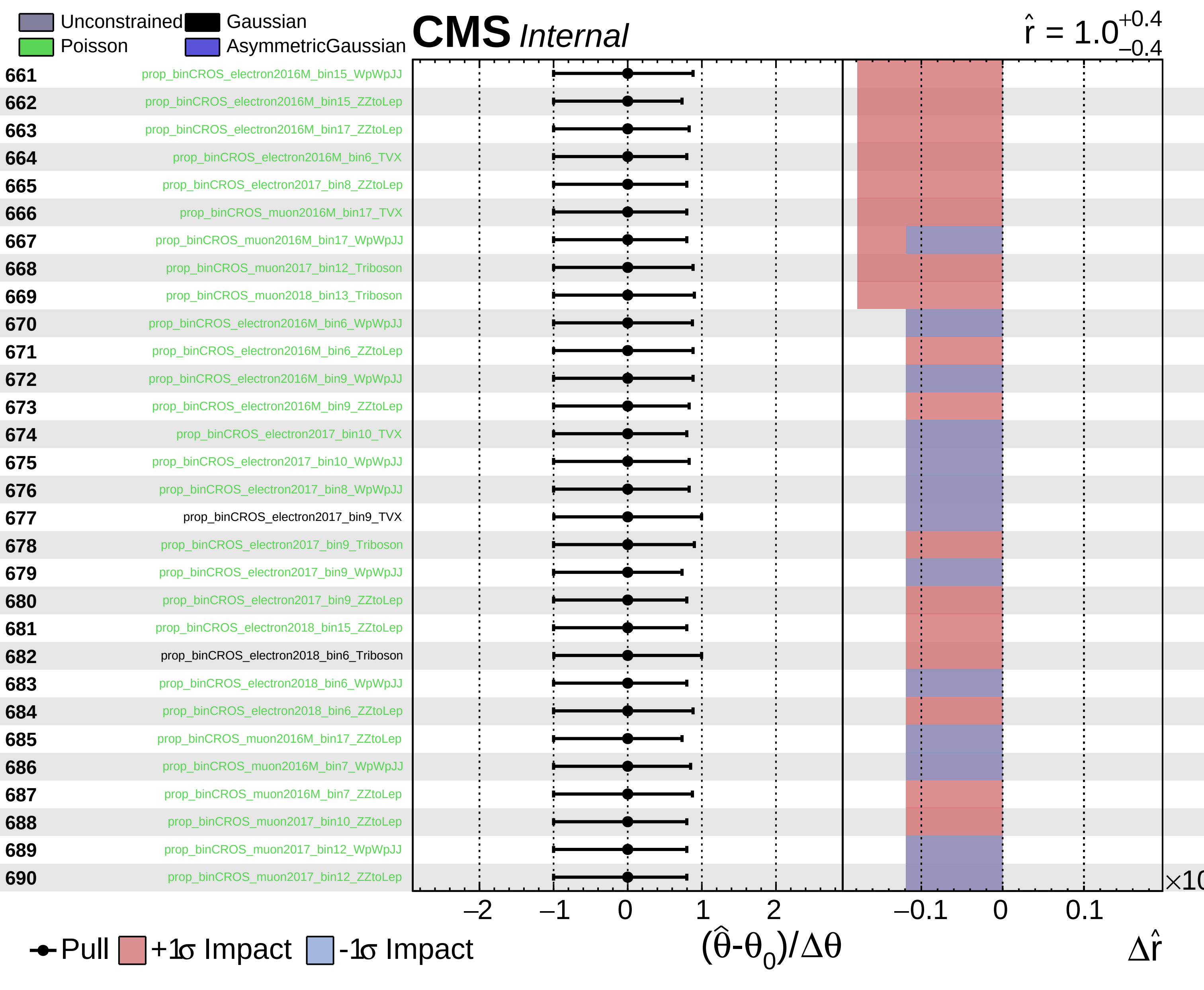
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.4}_{-0.4}$
541 prop_binCRF_electron2016M_bin15
542 prop_binCRTT_electron2016M_bin10
543 prop_binCROS_muon2017_bin14
544 prop_binCRF_electron2017_bin16
545 prop_binSR_electron2017_bin13_TVX
546 prop_binSR_electron2017_bin15_TVX
547 prop_binCROS_electron2018_bin6_TTTo2L2Nu
548 prop_binSR_muon2017_bin9_TVX
549 prop_binSR_electron2017_bin17_TVX
550 prop_binCROS_electron2018_bin13_Fake
551 prop_binCROS_electron2016M_bin9_Fake
552 prop_binCROS_muon2017_bin9_TTTo2L2Nu
553 prop_binCROS_muon2017_bin10_Fake
554 prop_binCROS_electron2017_bin10_TTTo2L2Nu
555 prop_binCROS_electron2018_bin15_TTTo2L2Nu
556 prop_binCRTT_muon2016M_bin11
557 prop_binCROS_muon2017_bin17_WrongSign
558 prop_binCROS_electron2016M_bin13
559 prop_binSR_muon2016M_bin15_Triboson
560 prop_binCROS_muon2016M_bin7_Fake
561 prop_binCROS_muon2016M_bin17_WrongSign
562 prop_binSR_muon2016M_bin16_TVX
563 prop_binCRTT_electron2017_bin5
564 prop_binCRF_electron2016M_bin12_TTTo2L2Nu
565 prop_binSR_electron2016M_bin2_ZZtoLep
566 prop_binCROS_electron2016M_bin15_TTTo2L2Nu
567 prop_binCROS_muon2017_bin10_TTTo2L2Nu
568 prop_binSR_electron2017_bin14_ZZtoLep
569 prop_binCRF_muon2017_bin12_TVX
570 prop_binSR_electron2017_bin15_ZZtoLep
−2 −1 0 1 2
−0.1 −0.05 0 0.05 0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



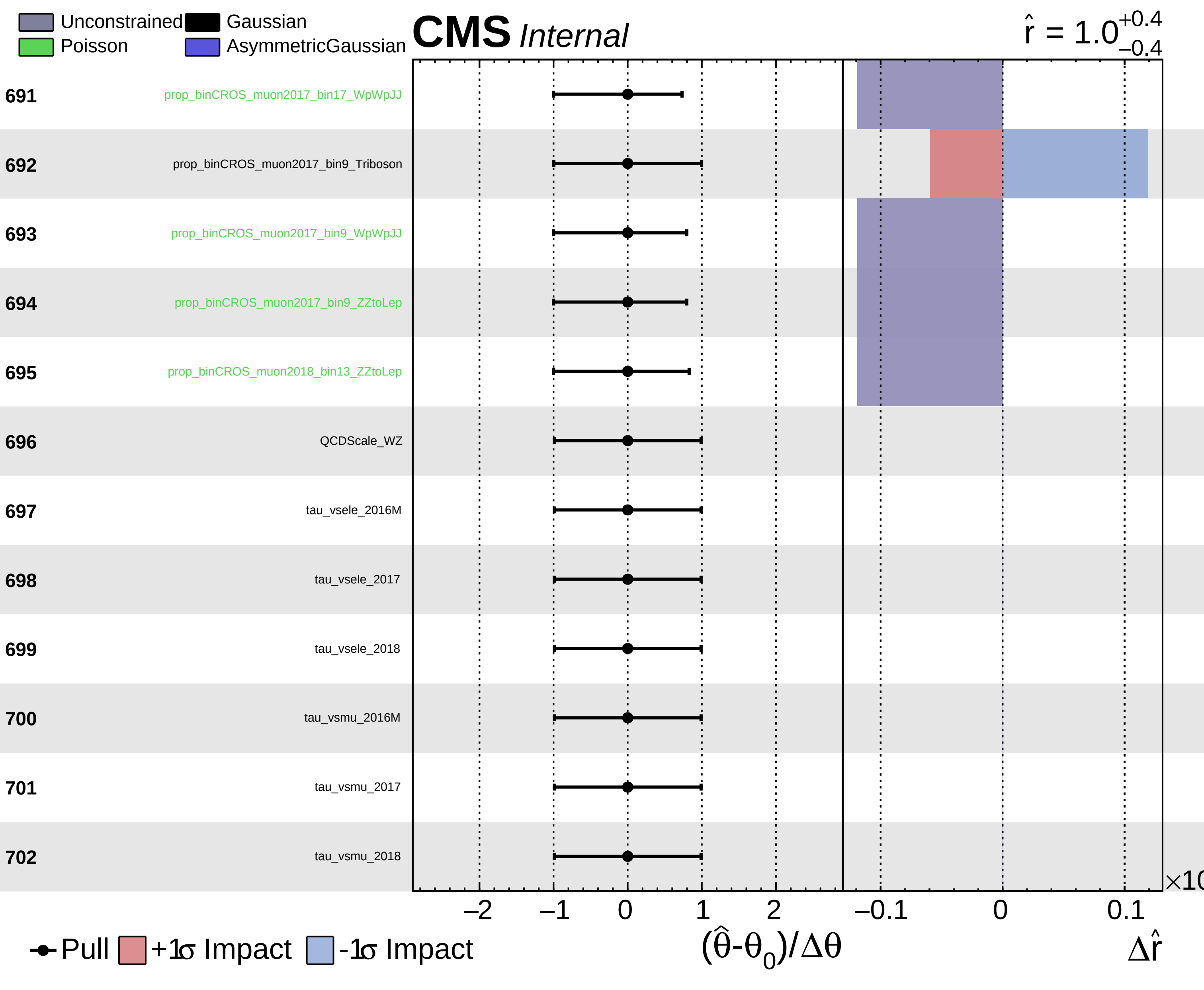
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.4}_{-0.4}$
571 prop_binCRF_electron2016M_bin10_WrongSign
572 prop_binCRF_muon2016M_bin11_Triboson
573 prop_binCROS_electron2018_bin6_VG
574 prop_binCROS_electron2017_bin17_TTTo2L2Nu
575 prop_binCROS_muon2017_bin17_TTTo2L2Nu
576 prop_binCROS_electron2016M_bin6_Fake
577 prop_binCROS_electron2016M_bin6_TTTo2L2Nu
578 prop_binCROS_electron2017_bin17_WrongSign
579 prop_binCROS_electron2016M_bin17_WrongSign
580 prop_binSR_muon2017_bin9_Triboson
581 prop_binCROS_muon2016M_bin17_TTTo2L2Nu
582 prop_binCROS_muon2017_bin12_TTTo2L2Nu
583 prop_binSR_electron2018_bin17_ZZtoLep
584 prop_binCROS_electron2018_bin17_TTTo2L2Nu
585 prop_binCROS_muon2016M_bin7_TTTo2L2Nu
586 prop_binCRF_electron2018_bin15_TVX
587 prop_binCROS_electron2017_bin9_TTTo2L2Nu
588 prop_binCRF_electron2018_bin15_Triboson
589 prop_binCROS_muon2018_bin17
590 prop_binSR_electron2016M_bin2_TVX
591 prop_binCROS_muon2017_bin9_VG
592 prop_binSR_electron2016M_bin2_WpWpJJ
593 prop_binSR_electron2016M_bin17_ZZtoLep
594 prop_binSR_electron2016M_bin16_ZZtoLep
595 prop_binCROS_electron2016M_bin17_TTTo2L2Nu
596 prop_binCROS_electron2017_bin9_WZ
597 prop_binCROS_electron2018_bin13_TTTo2L2Nu
598 prop_binCRF_muon2017_bin12_Triboson
599 prop_binSR_muon2016M_bin12_ZZtoLep
600 prop_binSR_electron2017_bin17_ZZtoLep
−2 −1 0 1 2
−0.04 −0.02 0 0.02 0.04
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.4}_{-0.4}$
601 prop_binCROS_electron2016M_bin16_WrongSign
602 prop_binSR_electron2016M_bin17_WrongSign
603 prop_binSR_electron2016M_bin16_WrongSign
604 prop_binCRF_electron2016M_bin10_Triboson
605 prop_binCRF_electron2016M_bin9_Triboson
606 prop_binSR_electron2016M_bin2_Triboson
607 prop_binSR_electron2017_bin15_WrongSign
608 prop_binSR_electron2016M_bin15_WrongSign
609 prop_binCROS_electron2016M_bin16_TTTo2L2Nu
610 prop_binCRF_electron2016M_bin12_WpWpJJ
611 prop_binSR_muon2017_bin16_WrongSign
612 prop_binCRF_electron2016M_bin9_ZZtoLep
613 prop_binCRF_electron2016M_bin10_ZZtoLep
614 prop_binCROS_muon2017_bin10_TVX
615 prop_binCROS_electron2017_bin17_Triboson
616 prop_binCROS_electron2016M_bin17_Triboson
617 prop_binCRF_electron2016M_bin12_Triboson
618 prop_binCRF_electron2017_bin12_Triboson
619 prop_binCROS_electron2018_bin17_Triboson
620 prop_binCROS_electron2016M_bin6_WZ
621 prop_binCROS_electron2017_bin17_TVX
622 prop_binCROS_muon2016M_bin17_Triboson
623 prop_binSR_muon2017_bin9_ZZtoLep
624 FES
625 prop_binCRF_electron2016M_bin12_TVX
626 prop_binCROS_electron2016M_bin15_Triboson
627 prop_binCROS_electron2018_bin13_Triboson
628 prop_binCROS_muon2016M_bin7_TVX
629 prop_binCROS_muon2017_bin12_TVX
630 prop_binCROS_muon2017_bin17_TVX
−2 −1 0 1 2
−10 −5 0 5 10
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.4}_{-0.4}$
631 prop_binCROS_electron2016M_bin16_Triboson
632 prop_binCROS_electron2017_bin17_WpWpJJ
633 prop_binCROS_muon2017_bin17_Triboson
634 prop_binCROS_electron2017_bin10_Triboson
635 prop_binCROS_muon2017_bin10_Triboson
636 prop_binCROS_electron2016M_bin17_WpWpJJ
637 prop_binCROS_electron2018_bin15_TVX
638 prop_binCROS_electron2018_bin17_WpWpJJ
639 prop_binCROS_muon2017_bin9_TVX
640 prop_binCROS_muon2018_bin13_TVX
641 prop_binCROS_electron2016M_bin16_WpWpJJ
642 prop_binCROS_electron2017_bin8_TVX
643 prop_binCROS_electron2017_bin8_Triboson
644 prop_binCROS_electron2018_bin17_ZZtoLep
645 prop_binCRF_electron2016M_bin12_ZZtoLep
646 prop_binCROS_electron2016M_bin16_TVX
647 prop_binCROS_electron2016M_bin16_ZZtoLep
648 prop_binCROS_electron2016M_bin9_Triboson
649 prop_binCROS_electron2017_bin9_VG
650 prop_binCROS_electron2018_bin13_ZZtoLep
651 prop_binCROS_electron2018_bin15_Triboson
652 prop_binCROS_electron2016M_bin6_Triboson
653 prop_binCROS_electron2016M_bin6_VG
654 prop_binCROS_electron2016M_bin9_TVX
655 prop_binCROS_electron2018_bin13_TVX
656 prop_binCROS_electron2018_bin13_WpWpJJ
657 prop_binCROS_electron2018_bin17_TVX
658 prop_binCROS_electron2018_bin6_TVX
659 prop_binCROS_muon2016M_bin7_Triboson
660 prop_binCROS_electron2016M_bin15_TVX
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.4}_{-0.4}$
661 prop_binCROS_electron2016M_bin15_WpWpJJ
662 prop_binCROS_electron2016M_bin15_ZZtoLep
663 prop_binCROS_electron2016M_bin17_ZZtoLep
664 prop_binCROS_electron2016M_bin6_TVX
665 prop_binCROS_electron2017_bin8_ZZtoLep
666 prop_binCROS_muon2016M_bin17_TVX
667 prop_binCROS_muon2016M_bin17_WpWpJJ
668 prop_binCROS_muon2017_bin12_Triboson
669 prop_binCROS_muon2018_bin13_Triboson
670 prop_binCROS_electron2016M_bin6_WpWpJJ
671 prop_binCROS_electron2016M_bin6_ZZtoLep
672 prop_binCROS_electron2016M_bin9_WpWpJJ
673 prop_binCROS_electron2016M_bin9_ZZtoLep
674 prop_binCROS_electron2017_bin10_TVX
675 prop_binCROS_electron2017_bin10_WpWpJJ
676 prop_binCROS_electron2017_bin8_WpWpJJ
677 prop_binCROS_electron2017_bin9_TVX
678 prop_binCROS_electron2017_bin9_Triboson
679 prop_binCROS_electron2017_bin9_WpWpJJ
680 prop_binCROS_electron2017_bin9_ZZtoLep
681 prop_binCROS_electron2018_bin15_ZZtoLep
682 prop_binCROS_electron2018_bin6_Triboson
683 prop_binCROS_electron2018_bin6_WpWpJJ
684 prop_binCROS_electron2018_bin6_ZZtoLep
685 prop_binCROS_muon2016M_bin17_ZZtoLep
686 prop_binCROS_muon2016M_bin7_WpWpJJ
687 prop_binCROS_muon2016M_bin7_ZZtoLep
688 prop_binCROS_muon2017_bin10_ZZtoLep
689 prop_binCROS_muon2017_bin12_WpWpJJ
690 prop_binCROS_muon2017_bin12_ZZtoLep
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 1.0 +0.4 −0.4
691 prop_binCROS_muon2017_bin17_WpWpJJ
692 prop_binCROS_muon2017_bin9_Triboson
693 prop_binCROS_muon2017_bin9_WpWpJJ
694 prop_binCROS_muon2017_bin9_ZZtoLep
695 prop_binCROS_muon2018_bin13_ZZtoLep
696 QCDScale_WZ
697 tau_vsele_2016M
698 tau_vsele_2017
699 tau_vsele_2018
700 tau_vsmu_2016M
701 tau_vsmu_2017
702 tau_vsmu_2018
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂